

Microsoft ADC Cybersecurity Skilling Program

Week 9 Assignment 17

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Securing Azure SQL Database

Introduction

In this lab, we will explore essential security configurations for protecting data stored in Azure SQL Databases as part of the AZ500: Microsoft Azure Security Technologies course. The primary objective is to understand and implement security best practices that ensure database integrity, confidentiality, and compliance. The lab focuses on deploying an Azure SQL Database and applying several layers of security controls. These include configuring **Advanced Data Protection** to guard against threats like SQL injection and data exfiltration, implementing **Data Classification** to discover and label sensitive information, and setting up **Auditing** to monitor and log database activities. By completing this lab, we will gain hands-on experience with securing SQL databases in the cloud, an essential skill for any Azure security professional.

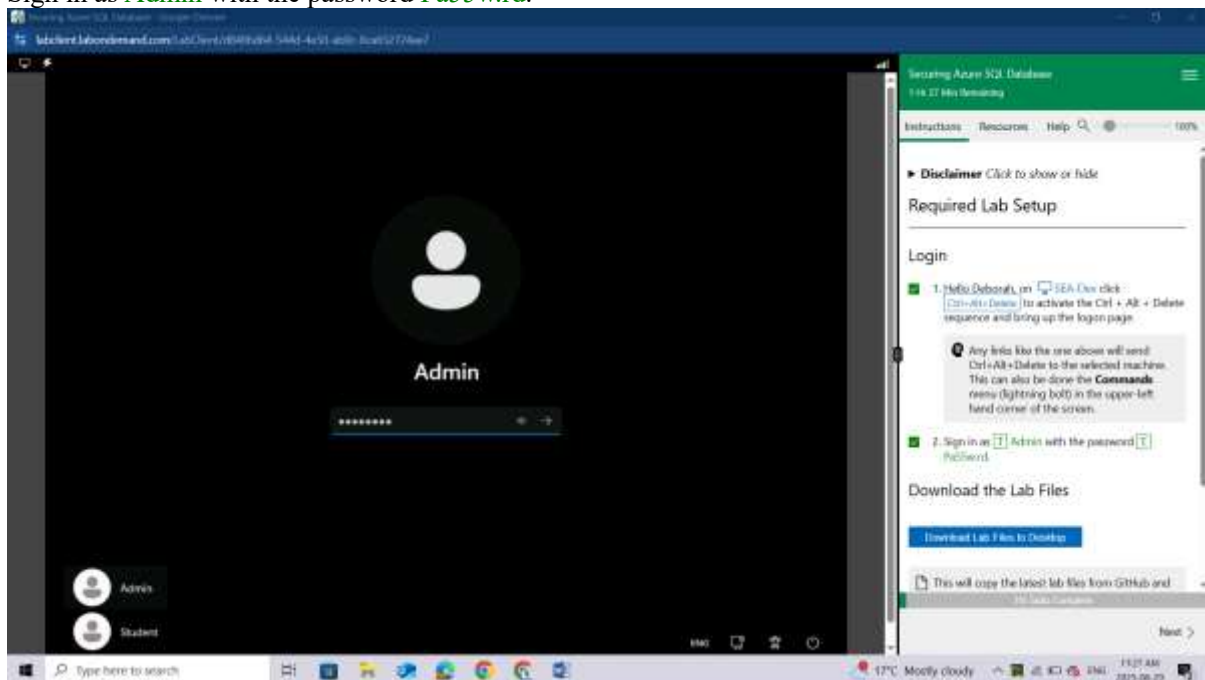
Tasks Completed

Required Lab Setup

Login:

Hello Deborah, on [SEA-Dev](#) click **Ctrl+Alt+Delete** to activate the Ctrl + Alt + Delete sequence and bring up the login page.

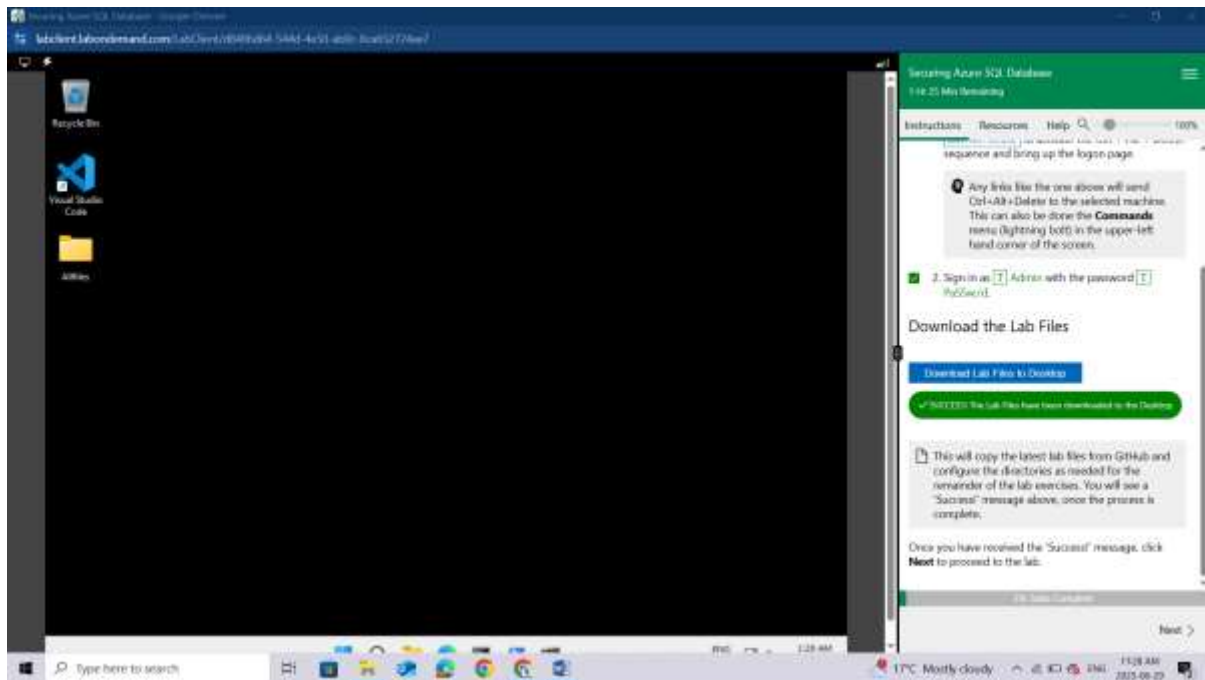
Sign in as **Admin** with the password **Pa55w.rd**.



Download the Lab Files

This will copy the latest lab files from GitHub and configure the directories as needed for the remainder of the lab exercises. You will see a 'Success!' message above, once the process is complete.

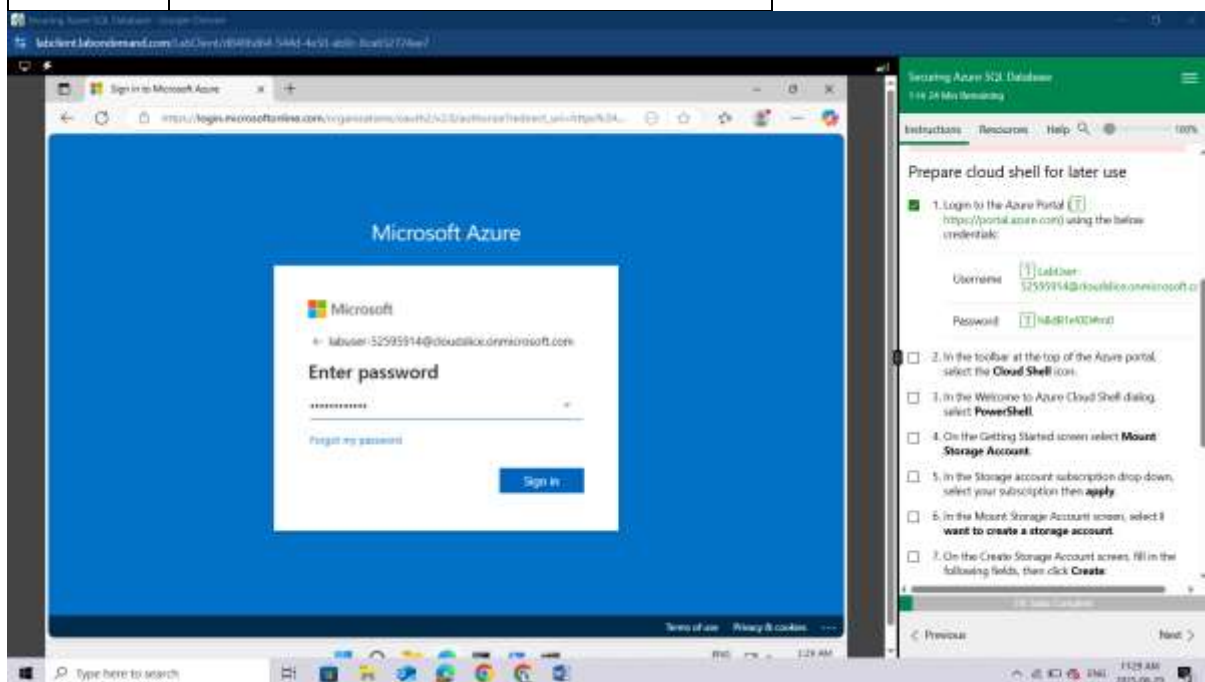
Once you have received the 'Success!' message, click **Next** to proceed to the lab.



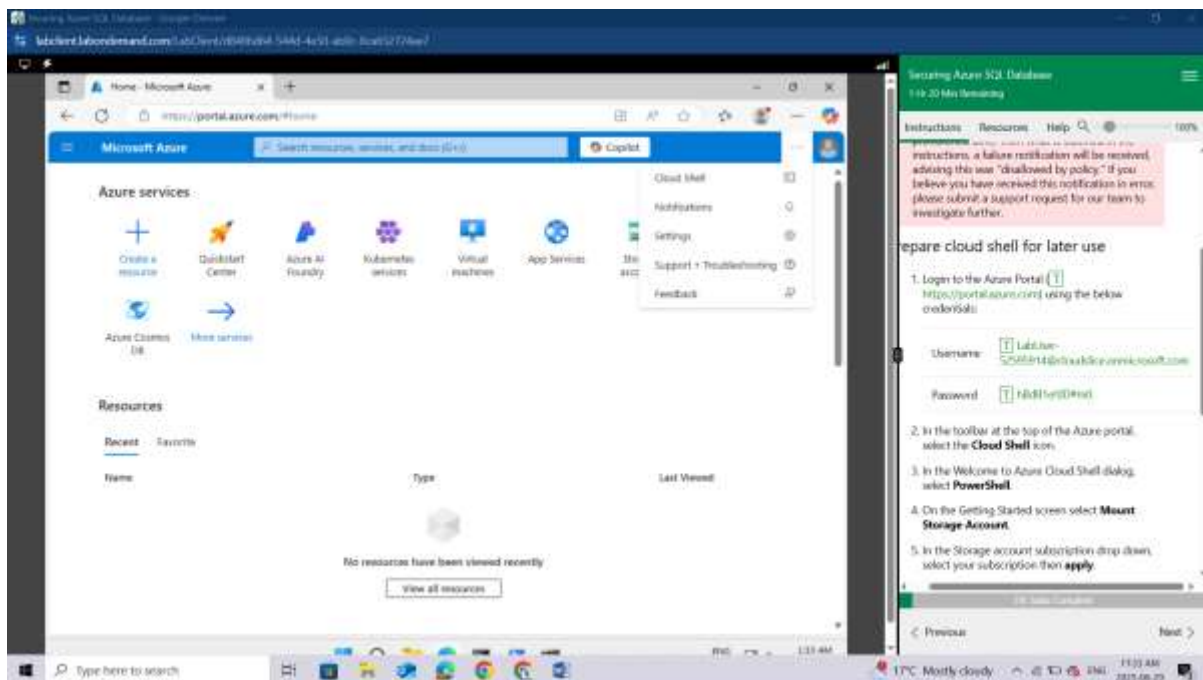
Prepare cloud shell for later use

Login to the Azure Portal (<https://portal.azure.com>) using the below credentials:

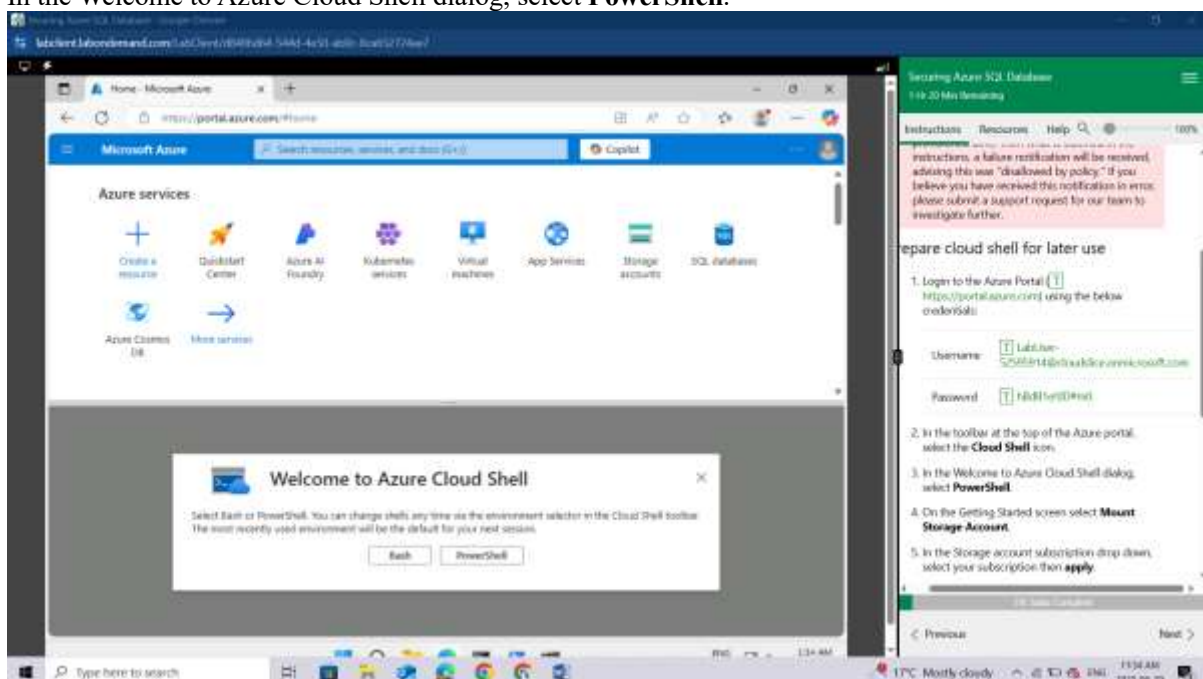
Username	LabUser-52595914@cloudslice.onmicrosoft.com
Password	hBdB1e!0D#m0



In the toolbar at the top of the Azure portal, select the **Cloud Shell** icon.

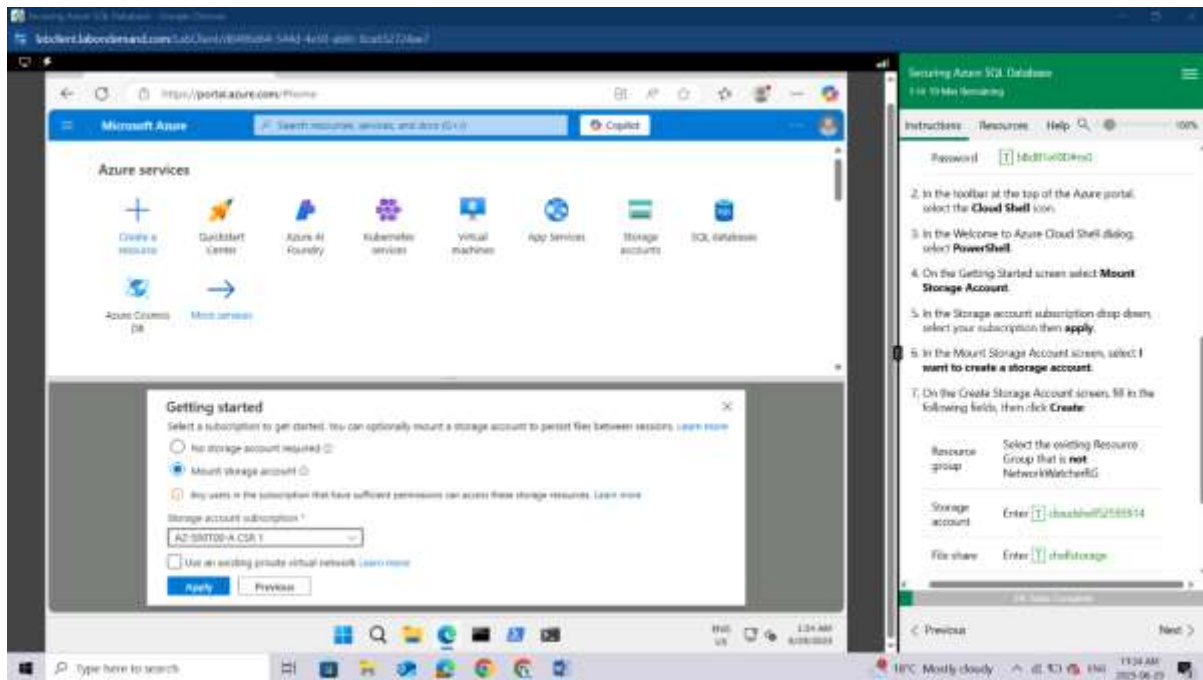


In the Welcome to Azure Cloud Shell dialog, select **PowerShell**.

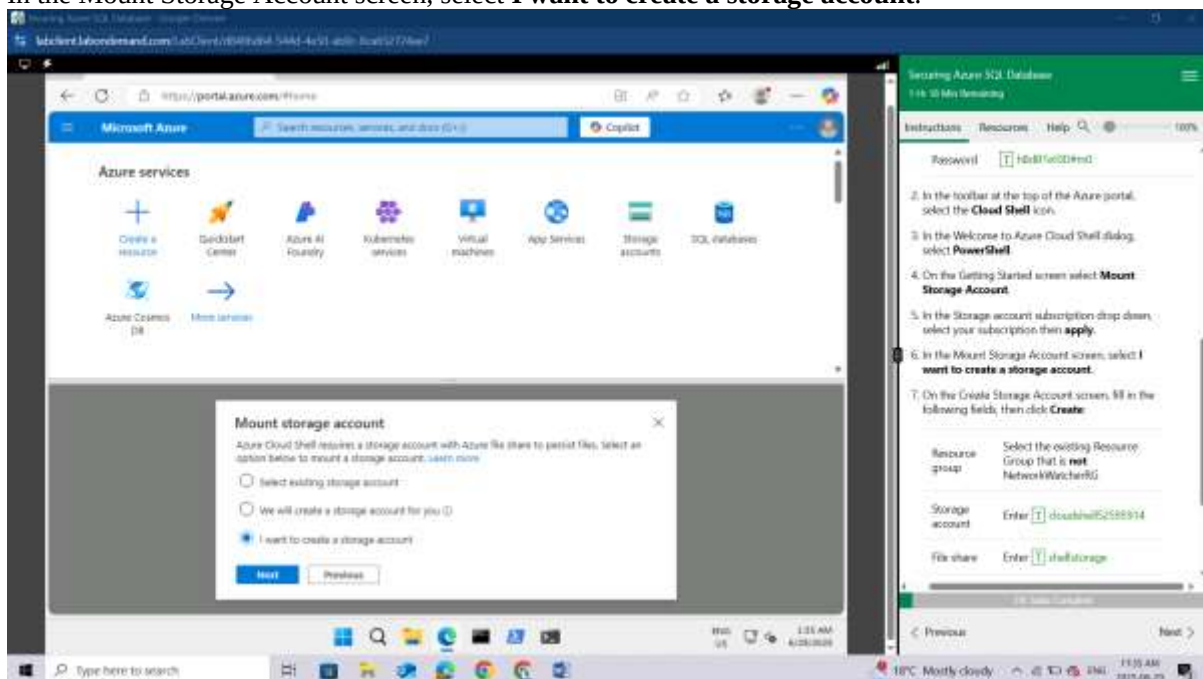


On the Getting Started screen select **Mount Storage Account**.

In the Storage account subscription drop down, select your subscription then **apply**.



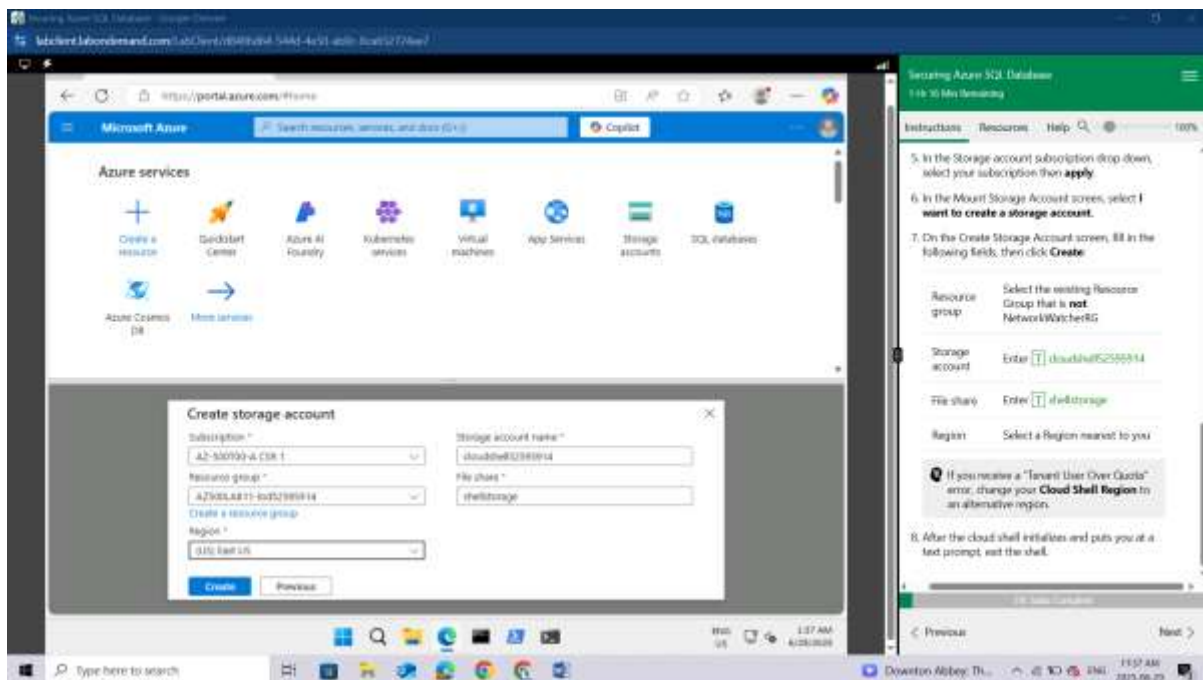
In the Mount Storage Account screen, select **I want to create a storage account**.



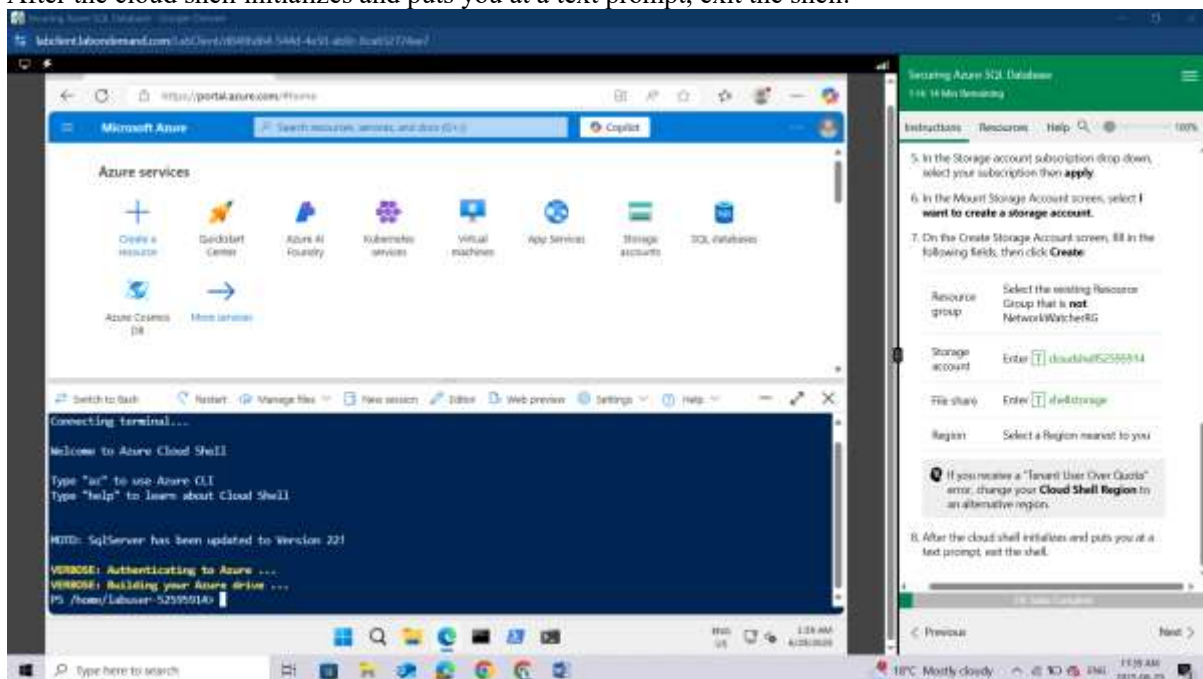
On the Create Storage Account screen, fill in the following fields, then click **Create**:

Resource group	Select the existing Resource Group that is not NetworkWatcherRG
Storage account	Enter cloudshell52595914
File share	Enter shellstorage
Region	Select a Region nearest to you

If you receive a "Tenant User Over Quota" error, change your **Cloud Shell Region** to an alternative region.



After the cloud shell initializes and puts you at a text prompt, exit the shell.



Lab 05: Securing Azure SQL Database

Student lab manual

Lab scenario

You have been asked to review security features for Azure SQL database. Specifically, you are interested in:

- Protection against attacks such as SQL injection and data exfiltration.
- Ability to discover and classify database information into categories such as Confidential.
- Ability to audit database server and database queries and log events.

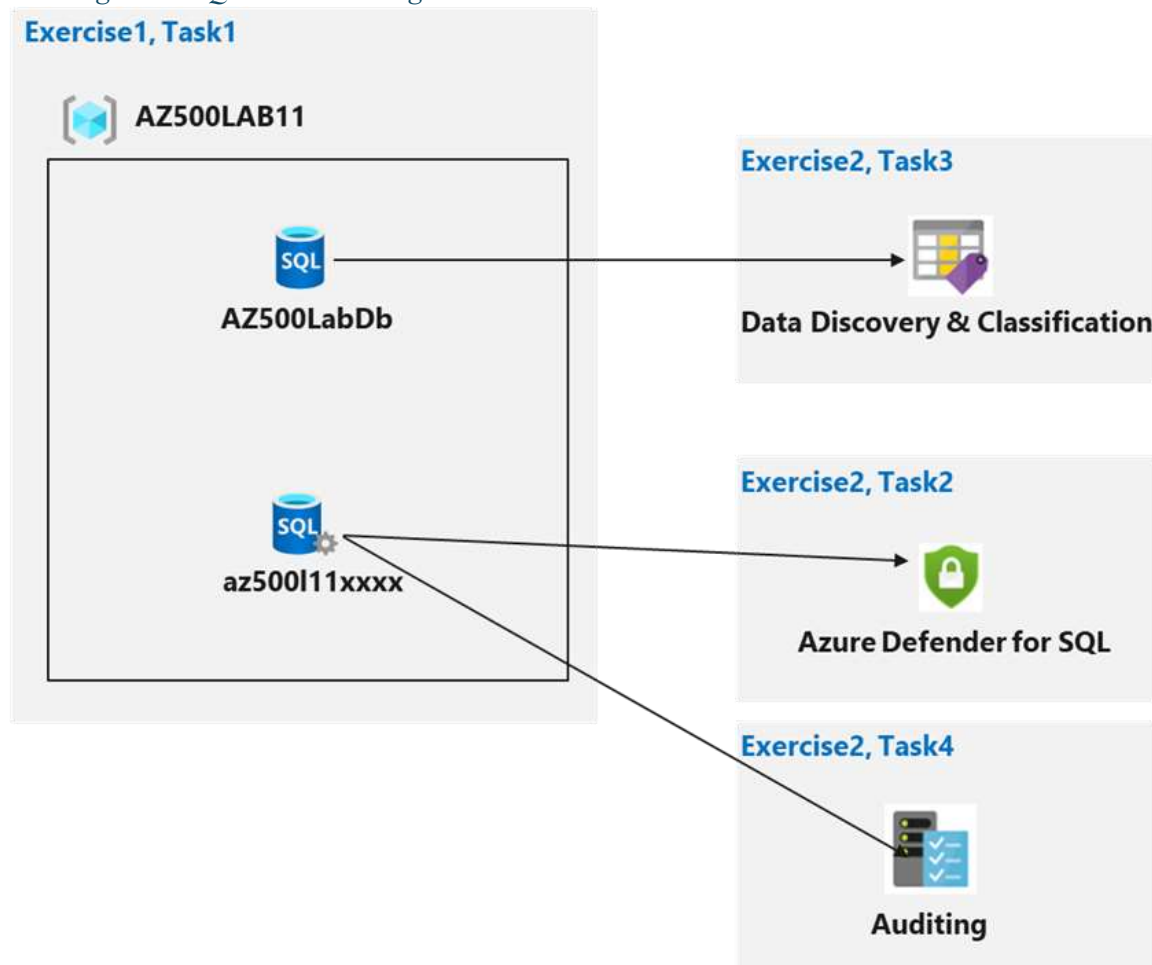
For all the resources in this lab, we are using the **East US** region. Verify with your instructor this is the region to use for class.

Lab objectives

In this lab, you will complete the following exercise:

- Exercise 1: Implement SQL Database security features

Securing Azure SQL Database diagram



Instructions

Lab files:

- \Allfiles\Labs\11\azuredeploy.json

Exercise 1: Implement SQL Database security features

Estimated timing: 30 minutes

In this exercise, you will complete the following tasks:

- Task 1: Deploy an Azure SQL Database
- Task 2: Configure Advanced Data Protection
- Task 3: Configure Data Classification
- Task 4: Configure Auditing

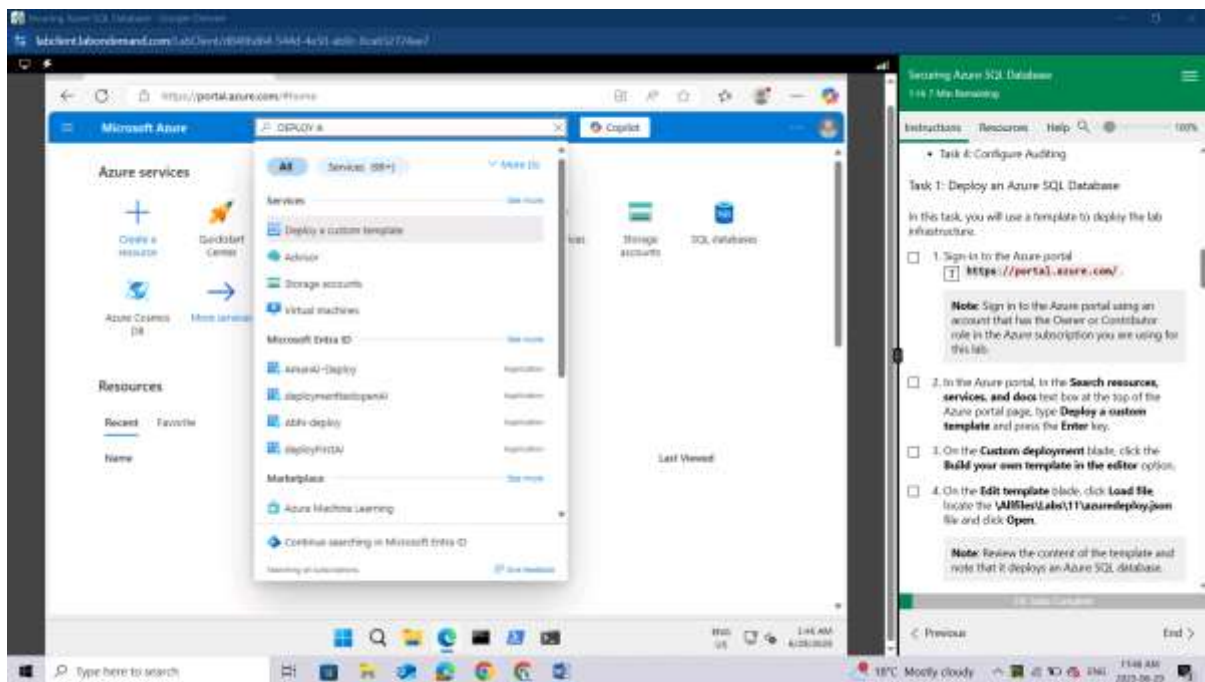
Task 1: Deploy an Azure SQL Database

In this task, you will use a template to deploy the lab infrastructure.

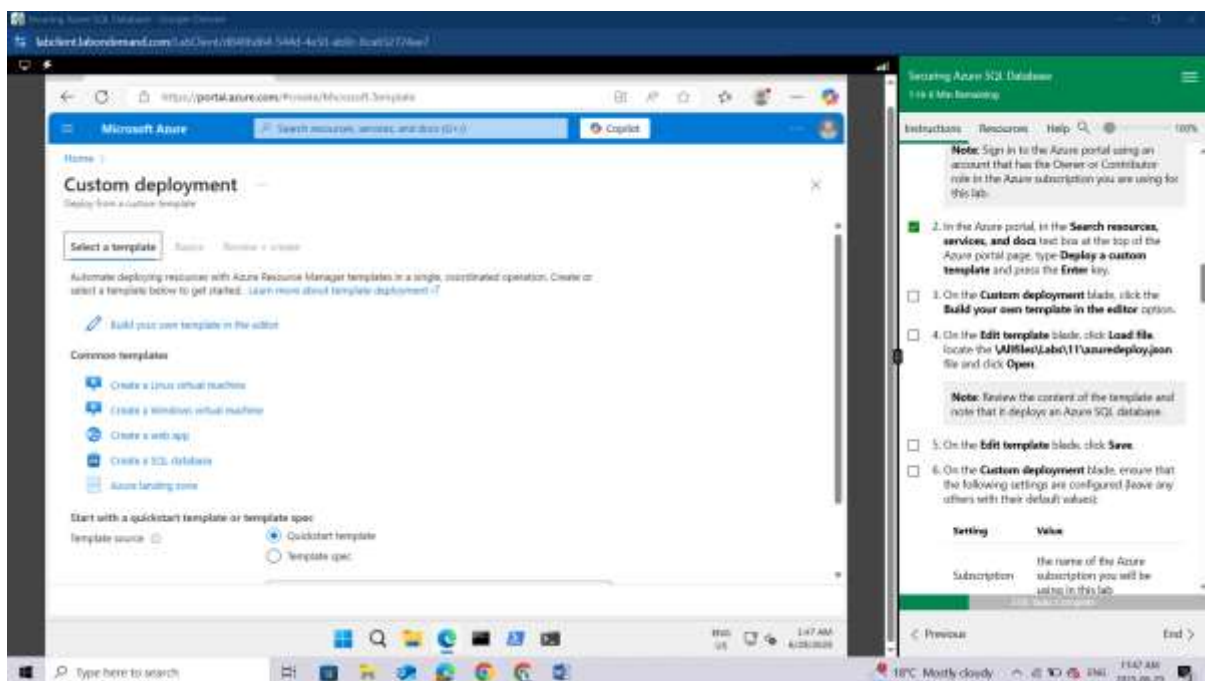
Sign-in to the Azure portal <https://portal.azure.com/>.

Note: Sign in to the Azure portal using an account that has the Owner or Contributor role in the Azure subscription you are using for this lab.

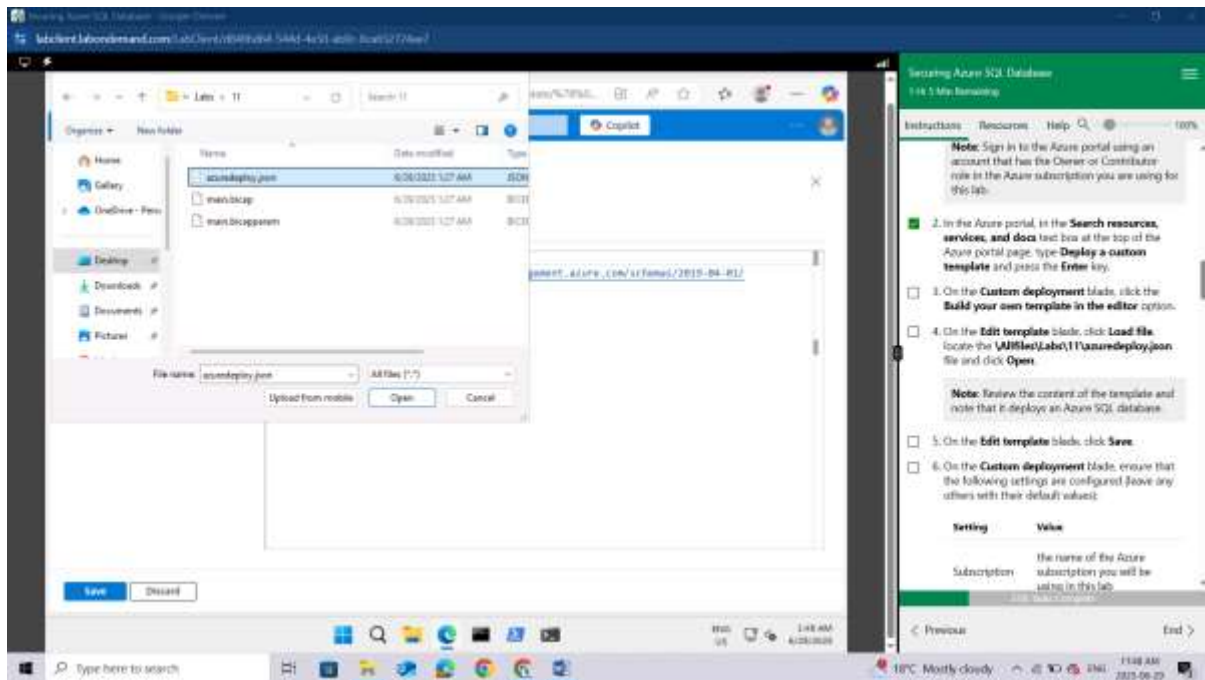
In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Deploy a custom template** and press the **Enter** key.



On the **Custom deployment** blade, click the **Build your own template in the editor** option.

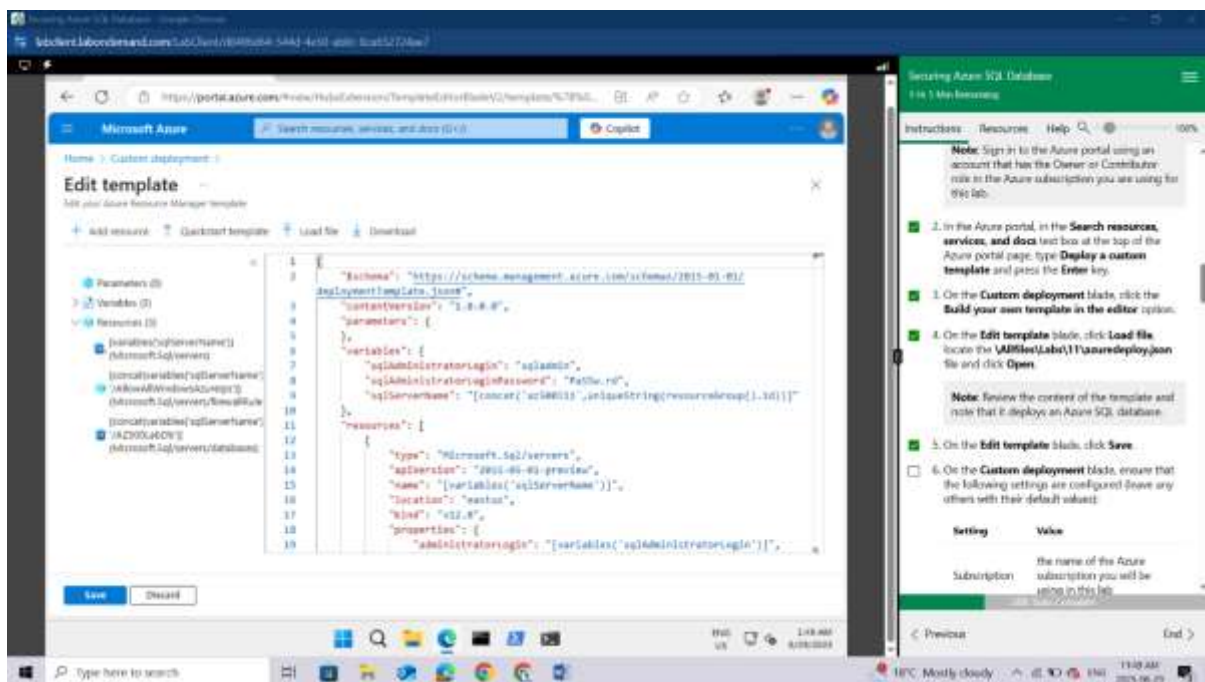


On the **Edit template** blade, click **Load file**, locate the **\Allfiles\Labs\11\azuredeploy.json** file and click **Open**.



Note: Review the content of the template and note that it deploys an Azure SQL database.

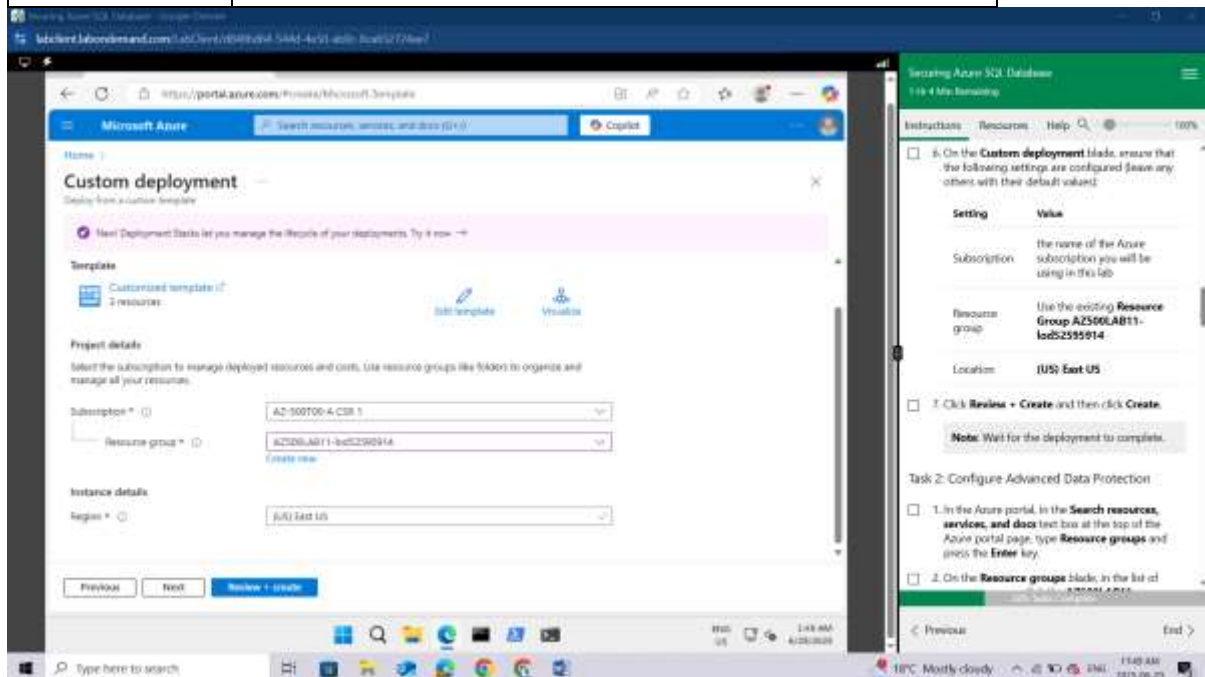
On the **Edit template** blade, click **Save**.



On the **Custom deployment** blade, ensure that the following settings are configured (leave any others with their default values):

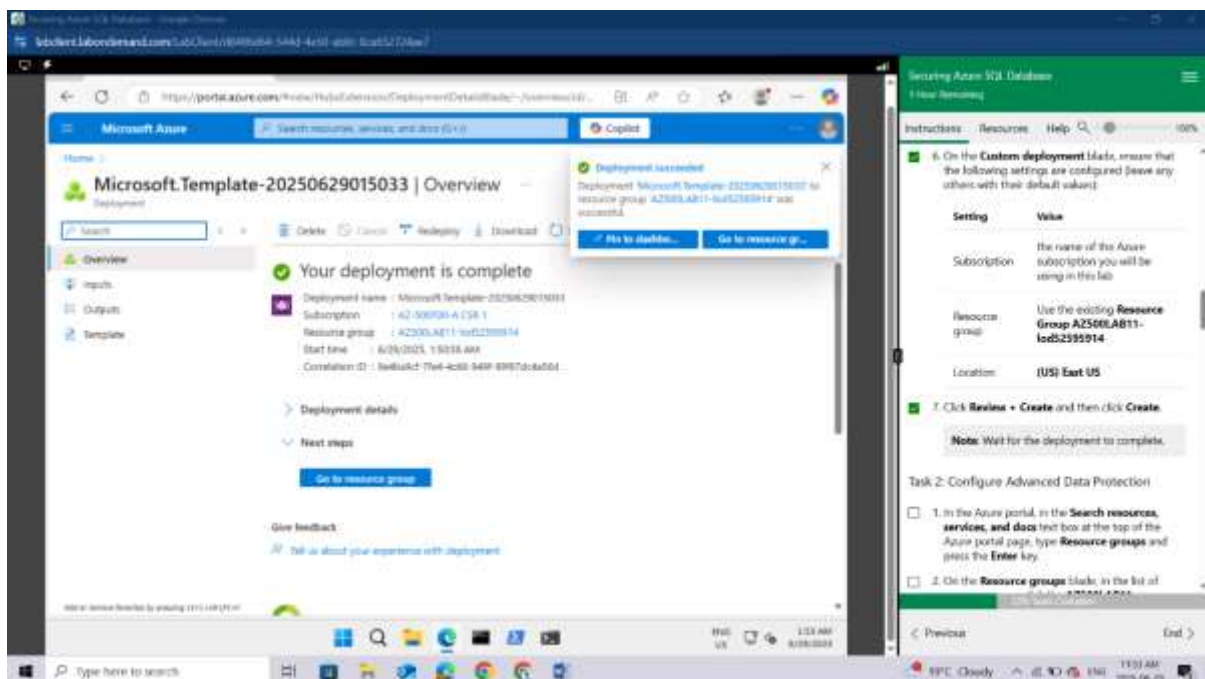
Setting	Value
Subscription	the name of the Azure subscription you will be using in this lab

Resource group	Use the existing Resource Group AZ500LAB11-lod52595914
Location	(US) East US



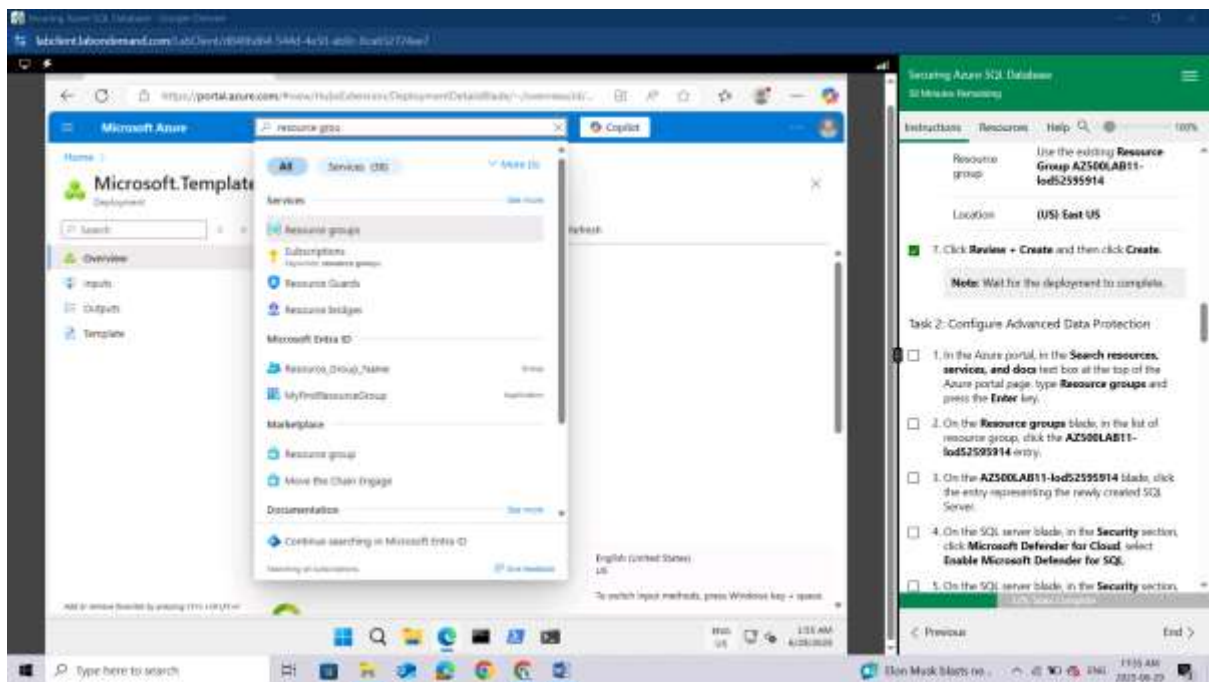
Click **Review + Create** and then click **Create**.

Note: Wait for the deployment to complete. It might take a while so you need to be patient

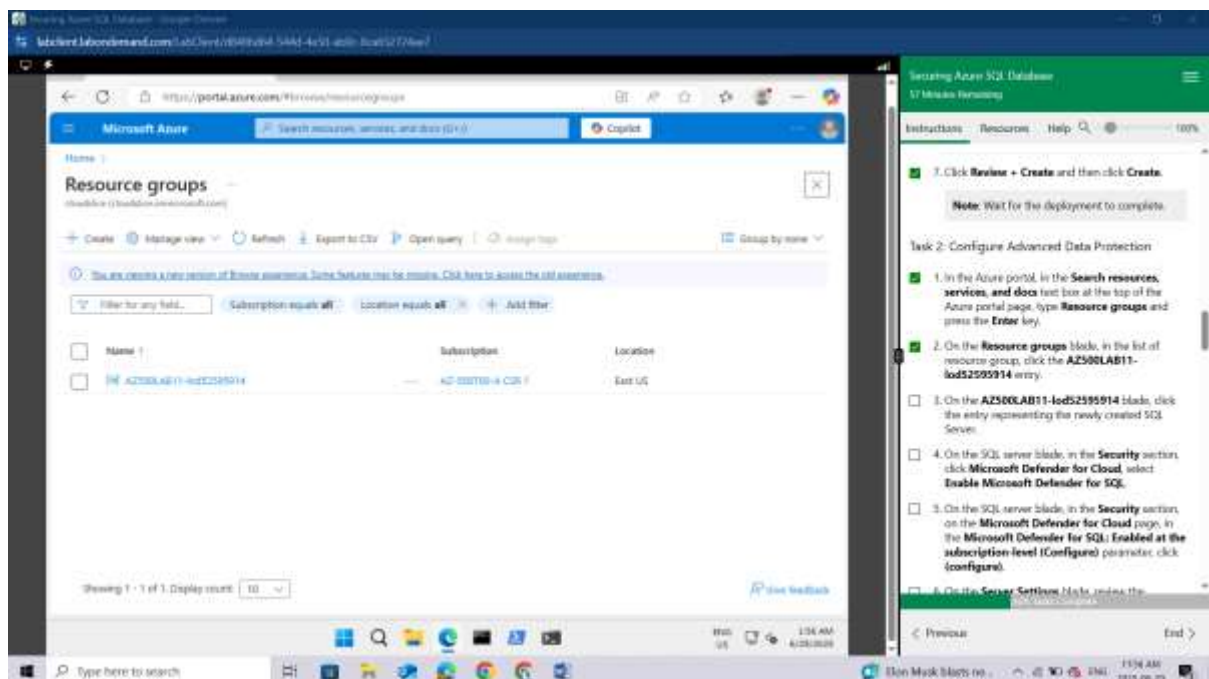


Task 2: Configure Advanced Data Protection

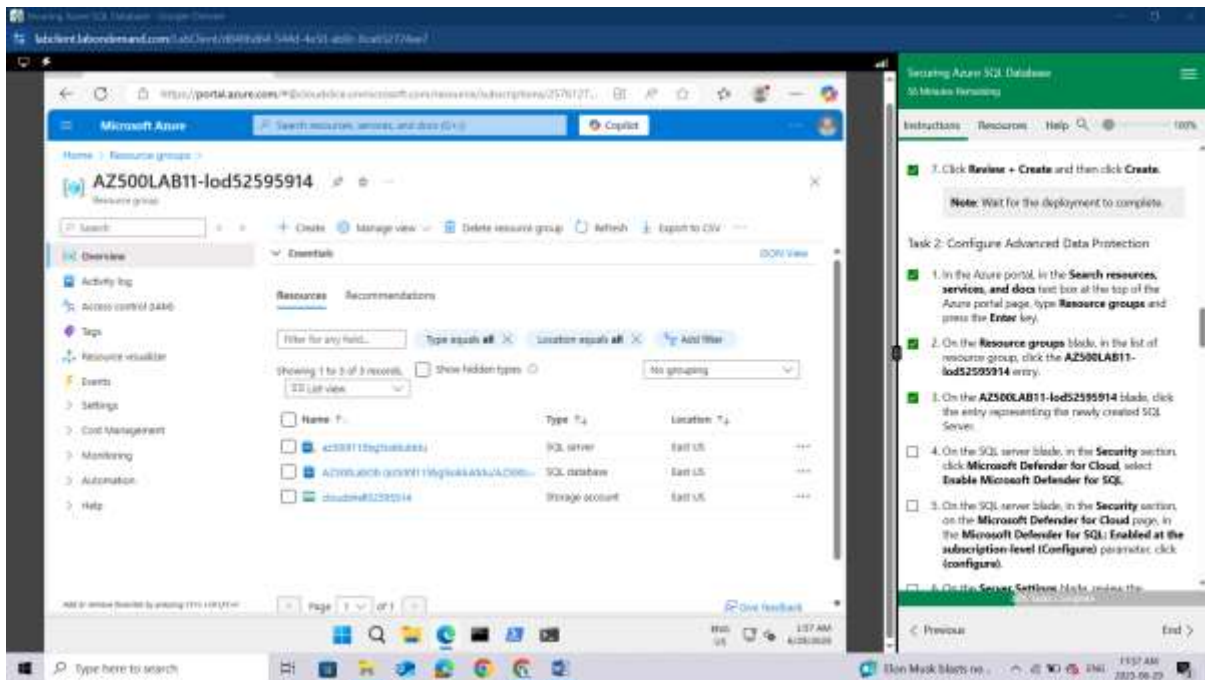
In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Resource groups** and press the **Enter** key.



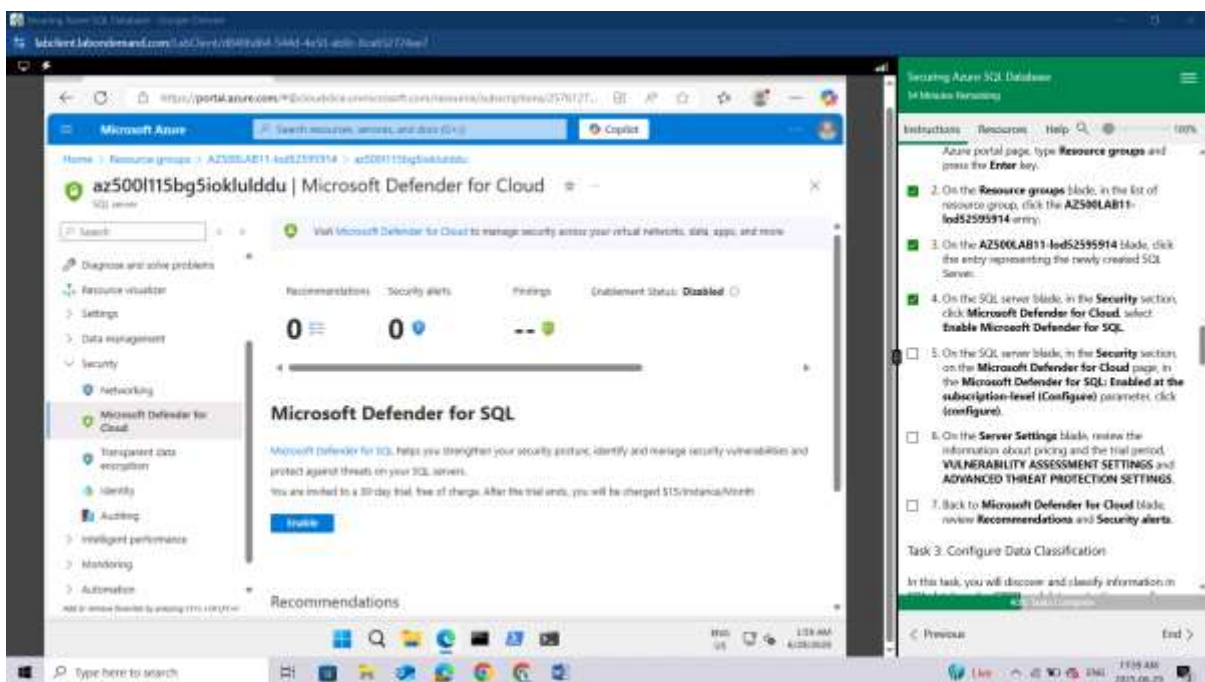
On the **Resource groups** blade, in the list of resource group, click the **AZ500LAB11-lod52595914** entry.



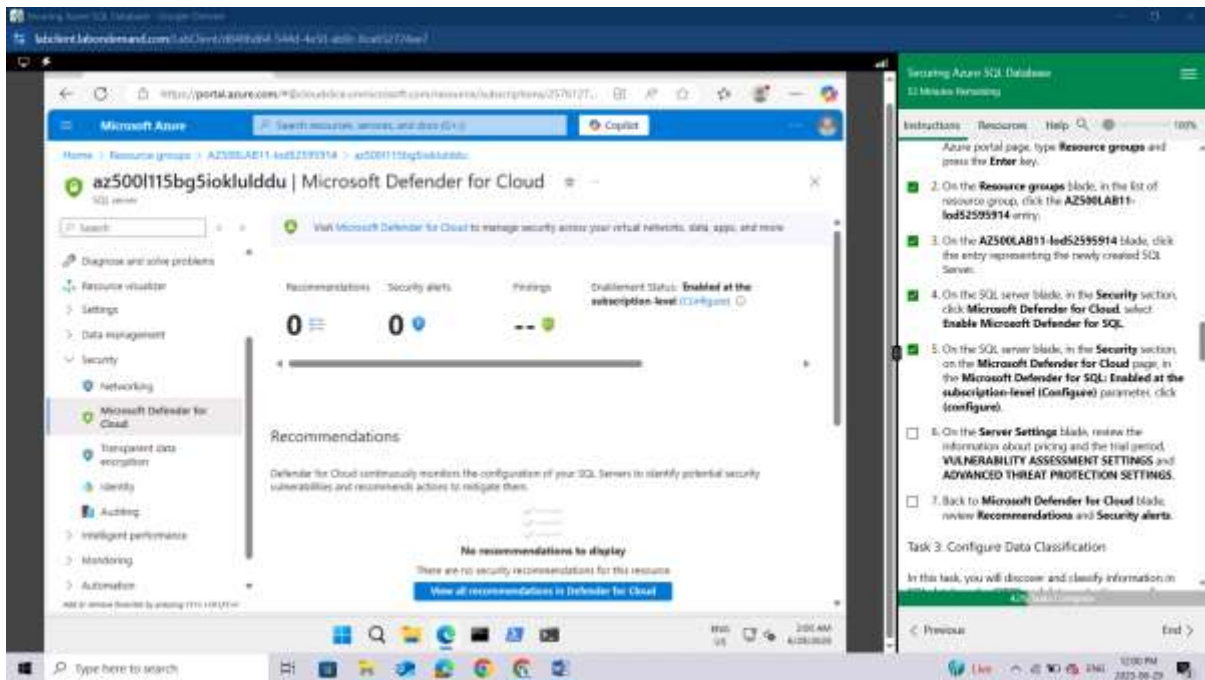
On the **AZ500LAB11-lod52595914** blade, click the entry representing the newly created SQL Server.



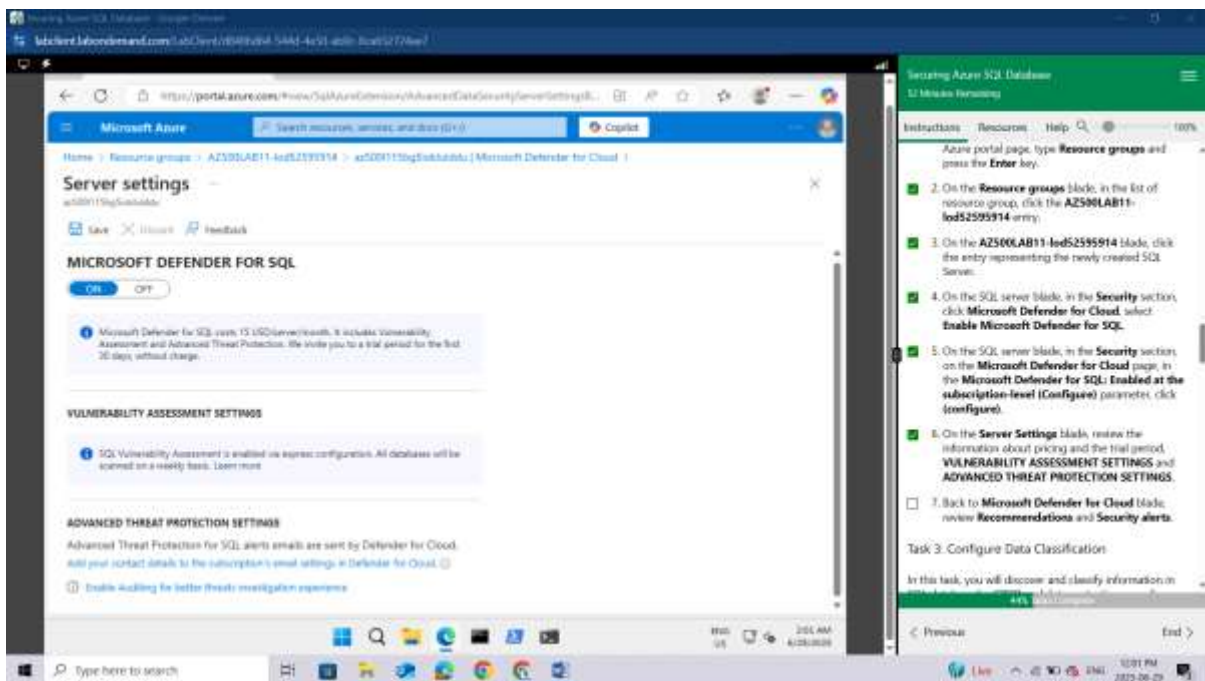
On the SQL server blade, in the **Security** section, click **Microsoft Defender for Cloud**, select **Enable Microsoft Defender for SQL**.



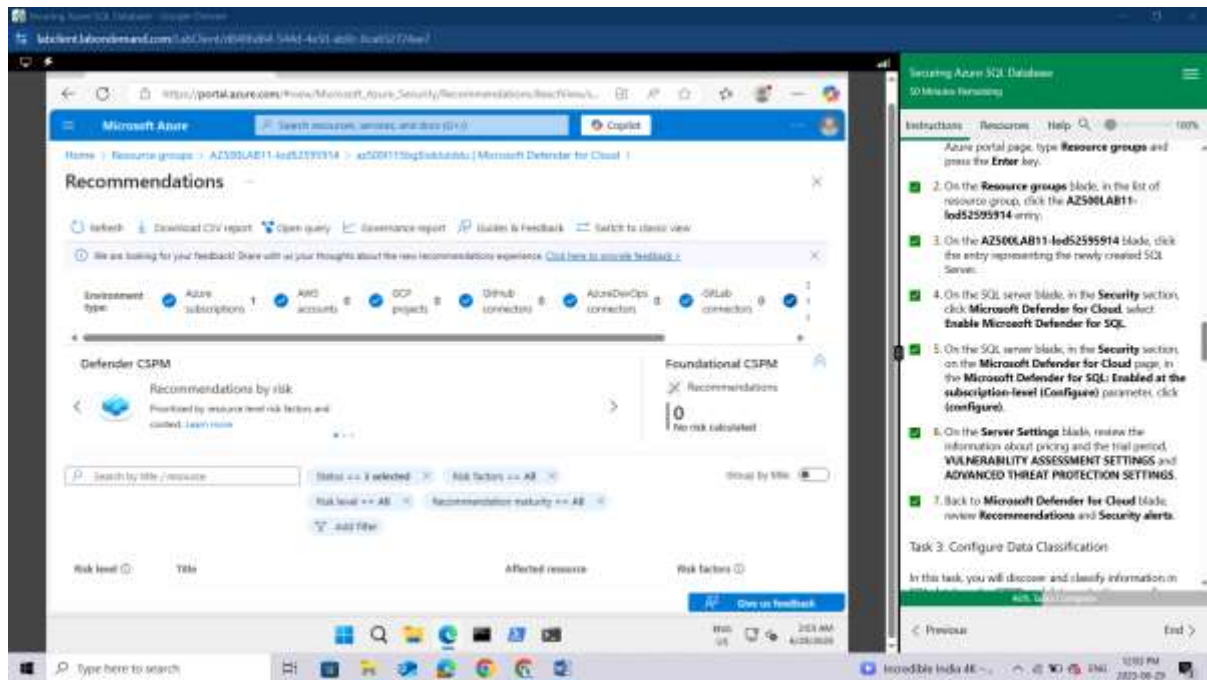
On the SQL server blade, in the **Security** section, on the **Microsoft Defender for Cloud** page, in the **Microsoft Defender for SQL: Enabled at the subscription-level (Configure)** parameter, click **(configure)**.



On the **Server Settings** blade, review the information about pricing and the trial period, **VULNERABILITY ASSESSMENT SETTINGS** and **ADVANCED THREAT PROTECTION SETTINGS**.



Back to **Microsoft Defender for Cloud** blade, review **Recommendations** and **Security alerts**.

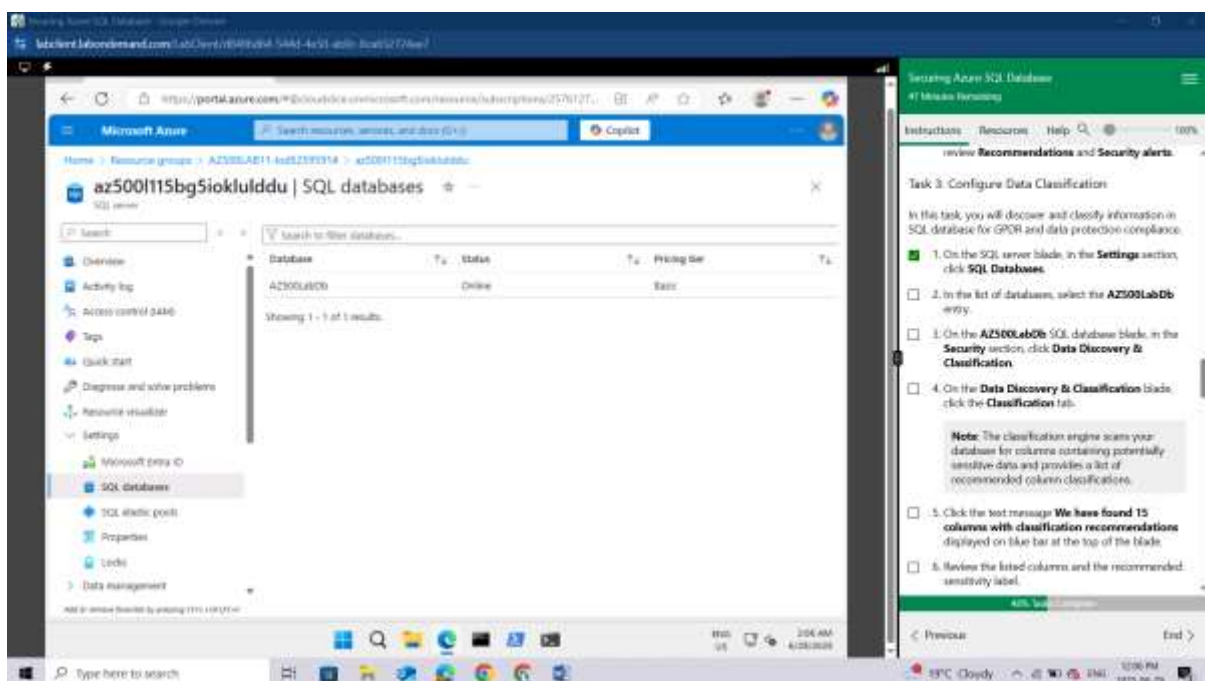


Task 3: Configure Data Classification

In this task, you will discover and classify information in SQL database for GDPR and data protection compliance.

On the SQL server blade, in the **Settings** section, click **SQL Databases**.

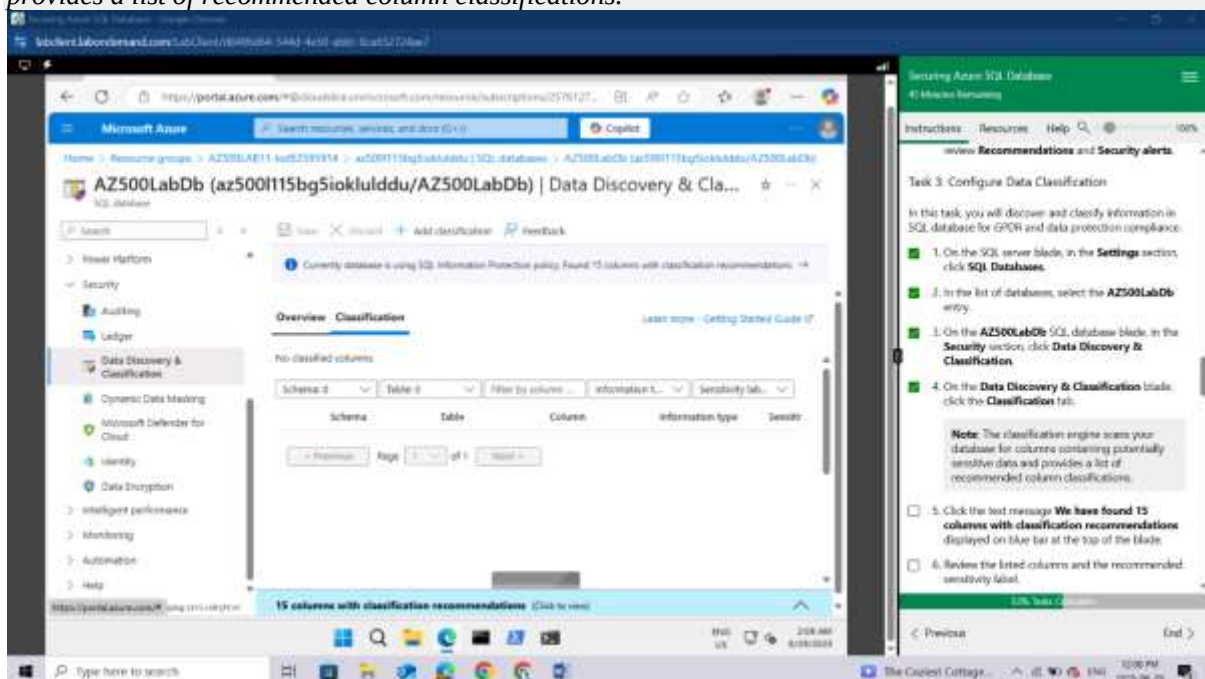
In the list of databases, select the **AZ500LabDb** entry.



On the **AZ500LabDb** SQL database blade, in the **Security** section, click **Data Discovery & Classification**.

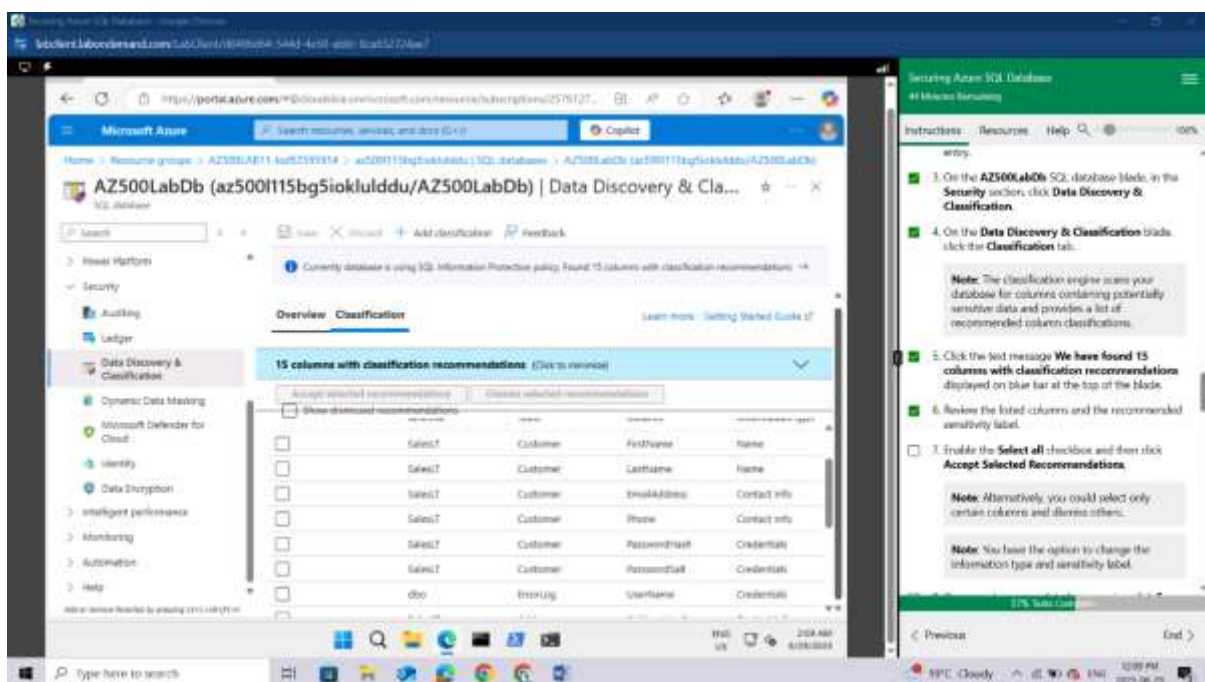
On the **Data Discovery & Classification** blade, click the **Classification** tab.

Note: The classification engine scans your database for columns containing potentially sensitive data and provides a list of recommended column classifications.



Click the text message **We have found 15 columns with classification recommendations** displayed on blue bar at the top of the blade.

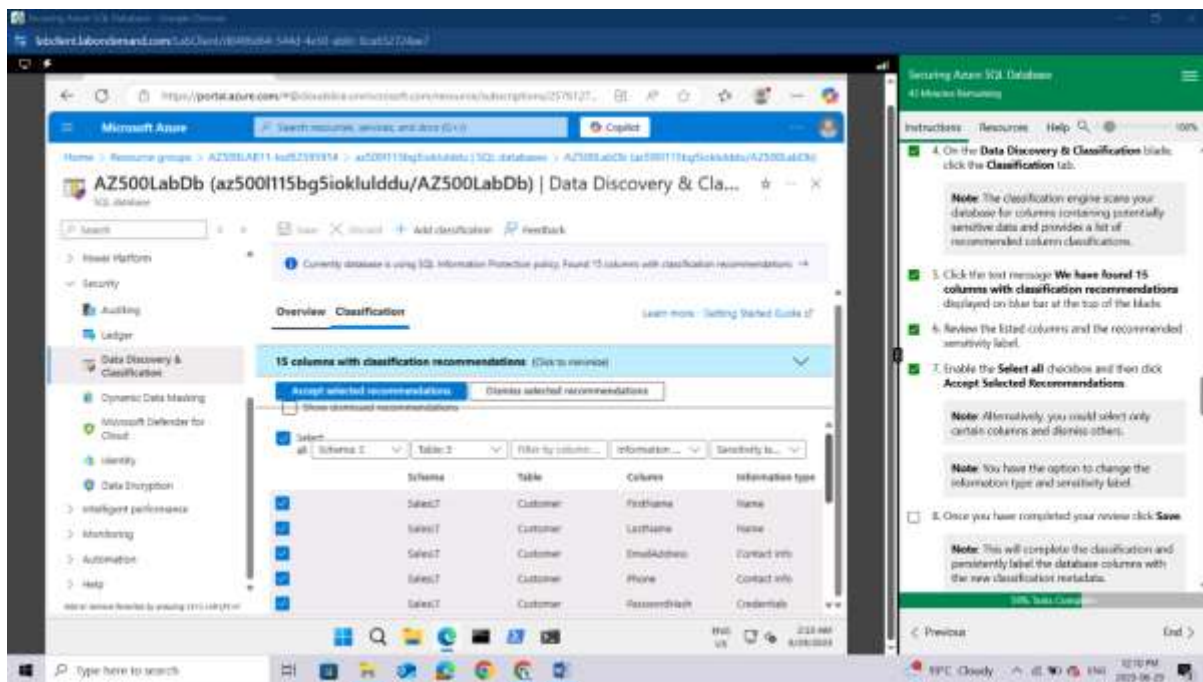
Review the listed columns and the recommended sensitivity label.



Enable the **Select all** checkbox and then click **Accept Selected Recommendations**.

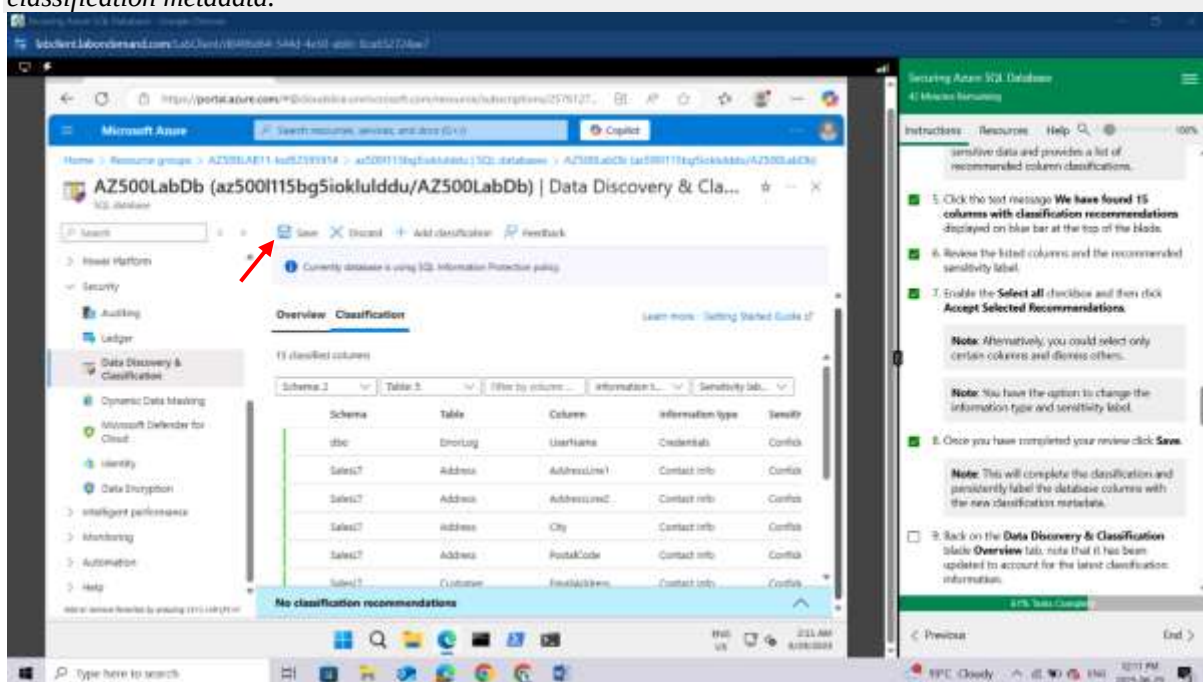
Note: Alternatively, you could select only certain columns and dismiss others.

Note: You have the option to change the information type and sensitivity label.

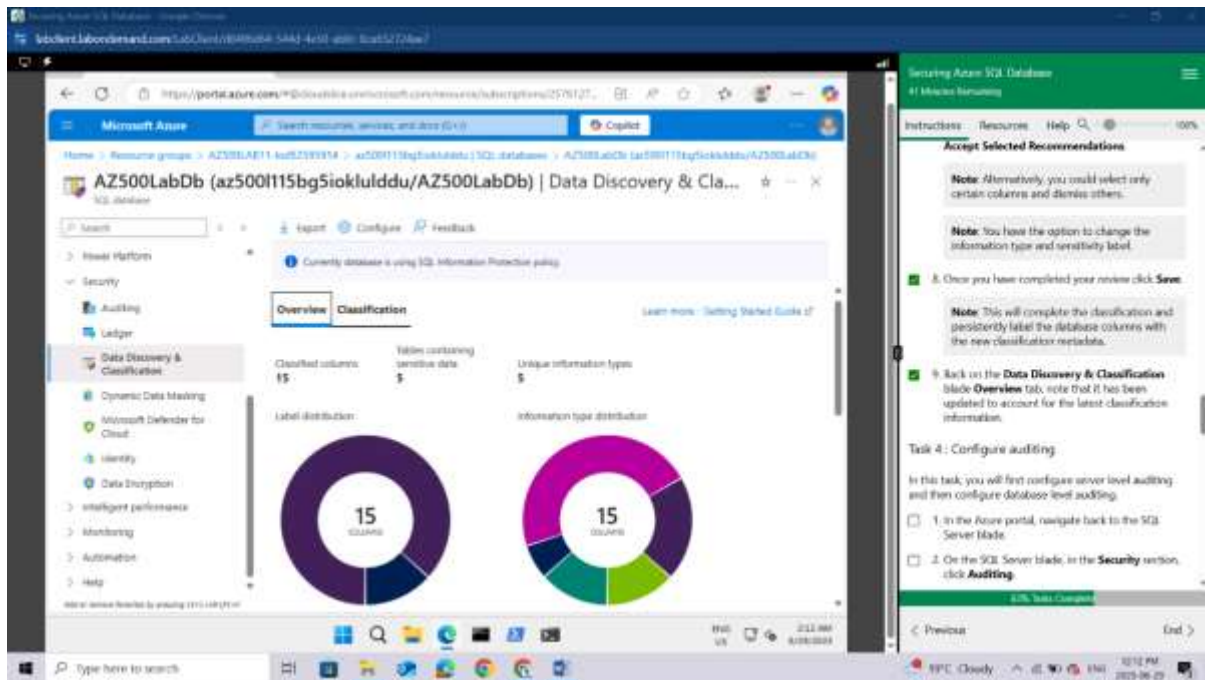


Once you have completed your review click **Save**.

Note: This will complete the classification and persistently label the database columns with the new classification metadata.



Back on the **Data Discovery & Classification** blade **Overview** tab, note that it has been updated to account for the latest classification information.



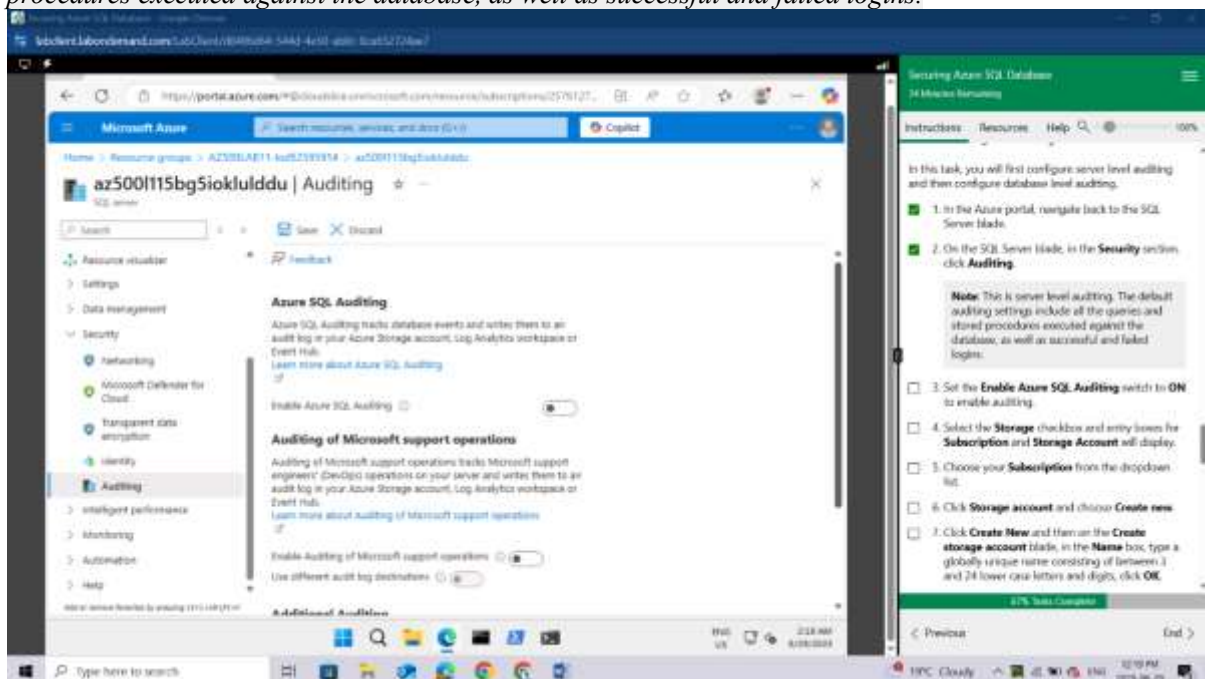
Task 4 : Configure auditing

In this task, you will first configure server level auditing and then configure database level auditing.

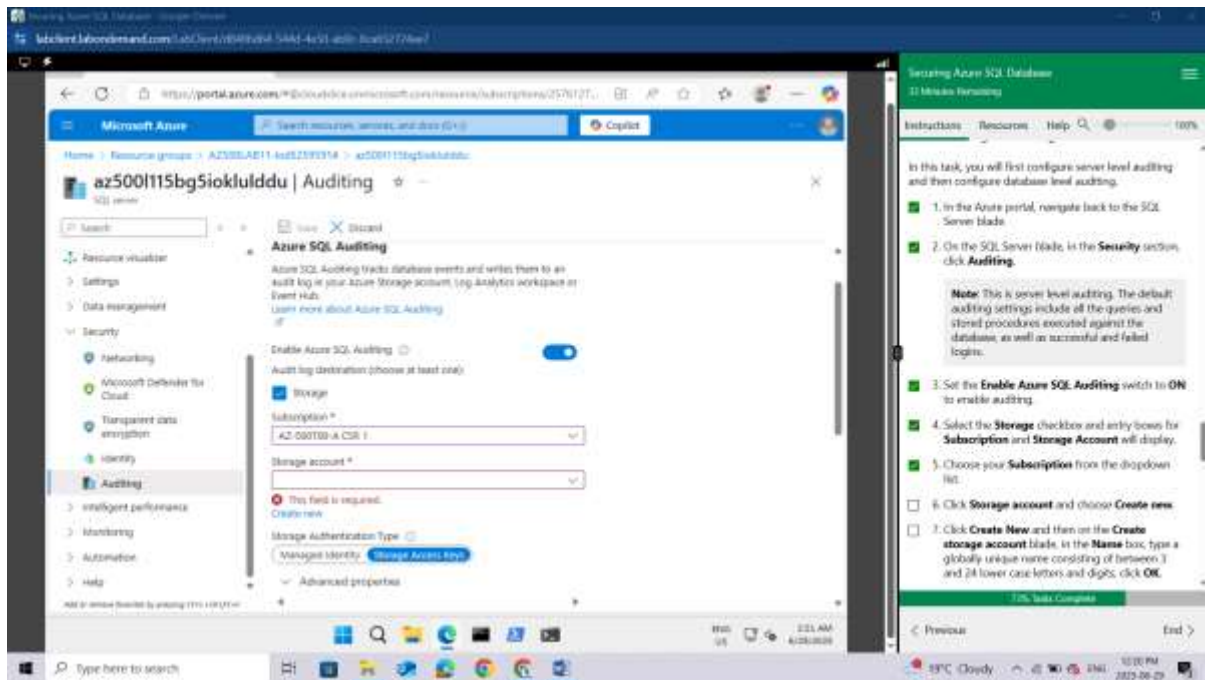
In the Azure portal, navigate back to the SQL Server blade.

On the SQL Server blade, in the **Security** section, click **Auditing**.

Note: This is server level auditing. The default auditing settings include all the queries and stored procedures executed against the database, as well as successful and failed logins.

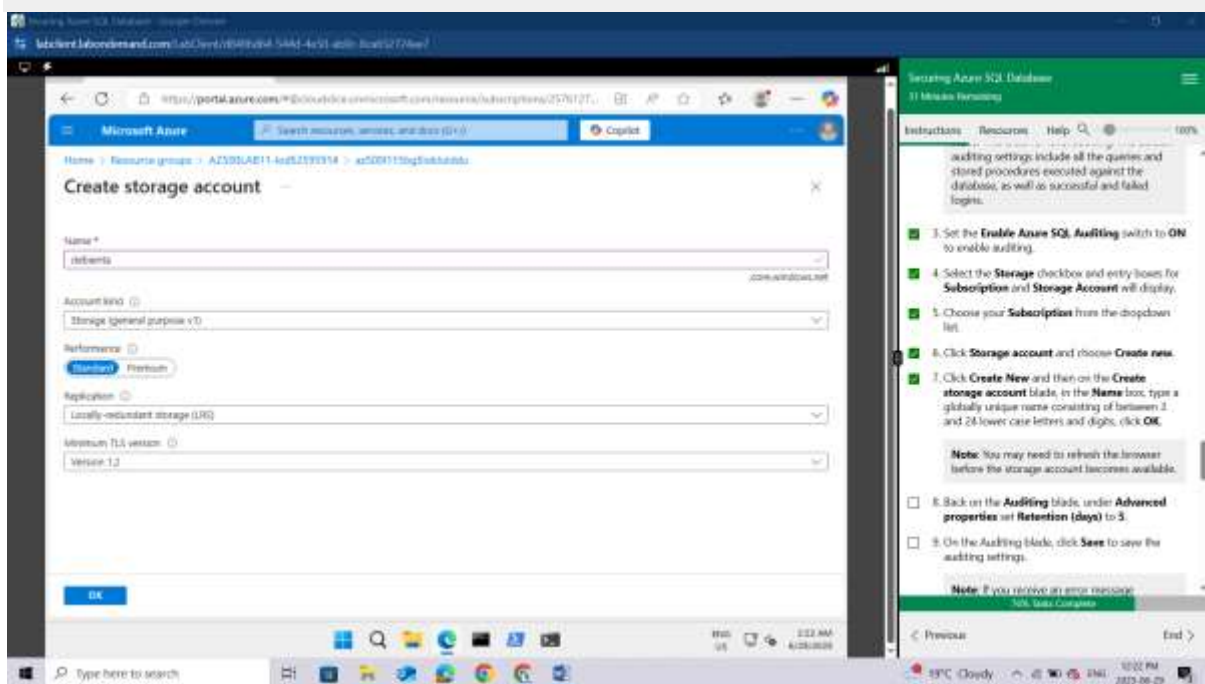


Set the **Enable Azure SQL Auditing** switch to **ON** to enable auditing. Select the **Storage** checkbox and entry boxes for **Subscription** and **Storage Account** will display. Choose your **Subscription** from the dropdown list.



Click **Storage account** and choose **Create new**. Click **Create New** and then on the **Create storage account** blade, in the **Name** box, type a globally unique name consisting of between 3 and 24 lower case letters and digits, click **OK**,

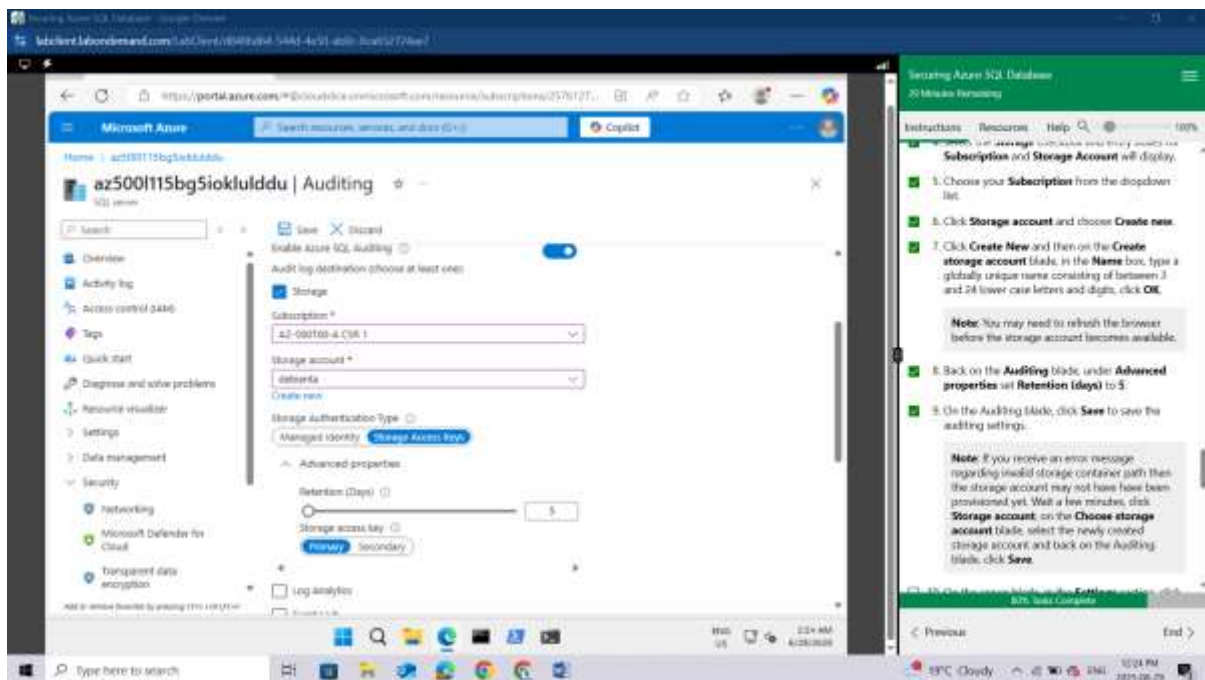
Note: You may need to refresh the browser before the storage account becomes available.



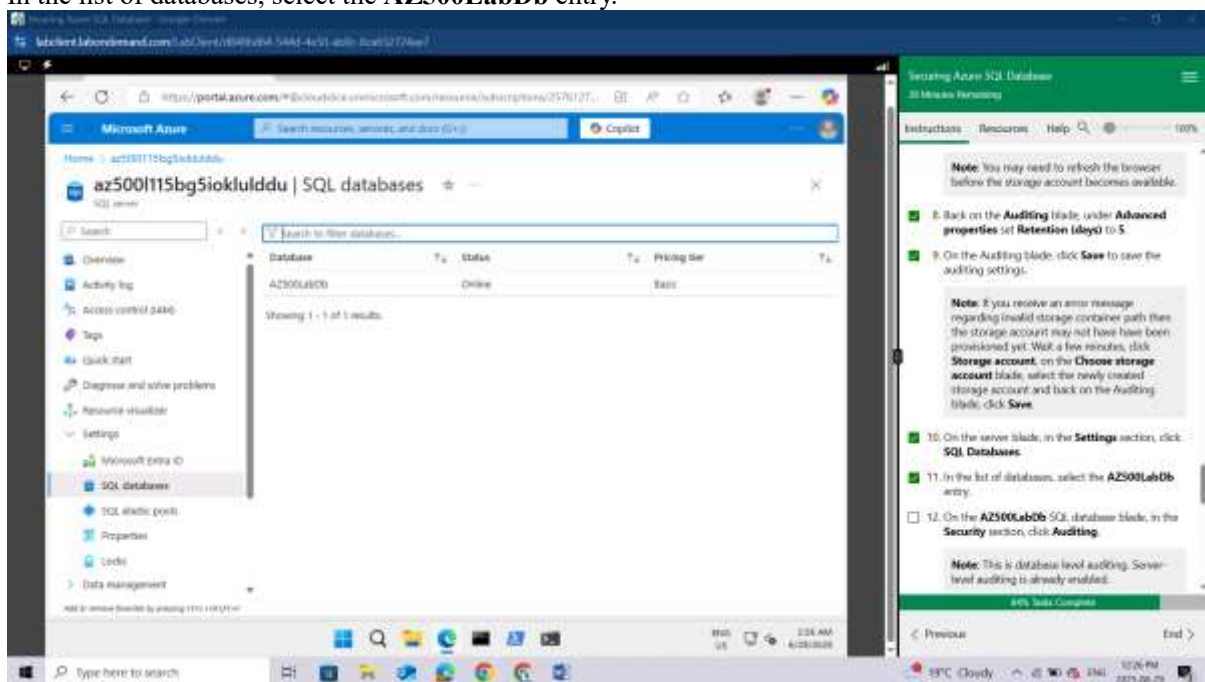
Back on the **Auditing** blade, under **Advanced properties** set **Retention (days)** to **5**.

On the Auditing blade, click **Save** to save the auditing settings.

Note: If you receive an error message regarding invalid storage container path then the storage account may not have been provisioned yet. Wait a few minutes, click **Storage account**, on the **Choose storage account** blade, select the newly created storage account and back on the Auditing blade, click **Save**.



On the server blade, in the **Settings** section, click **SQL Databases**.
In the list of databases, select the **AZ500LabDb** entry.

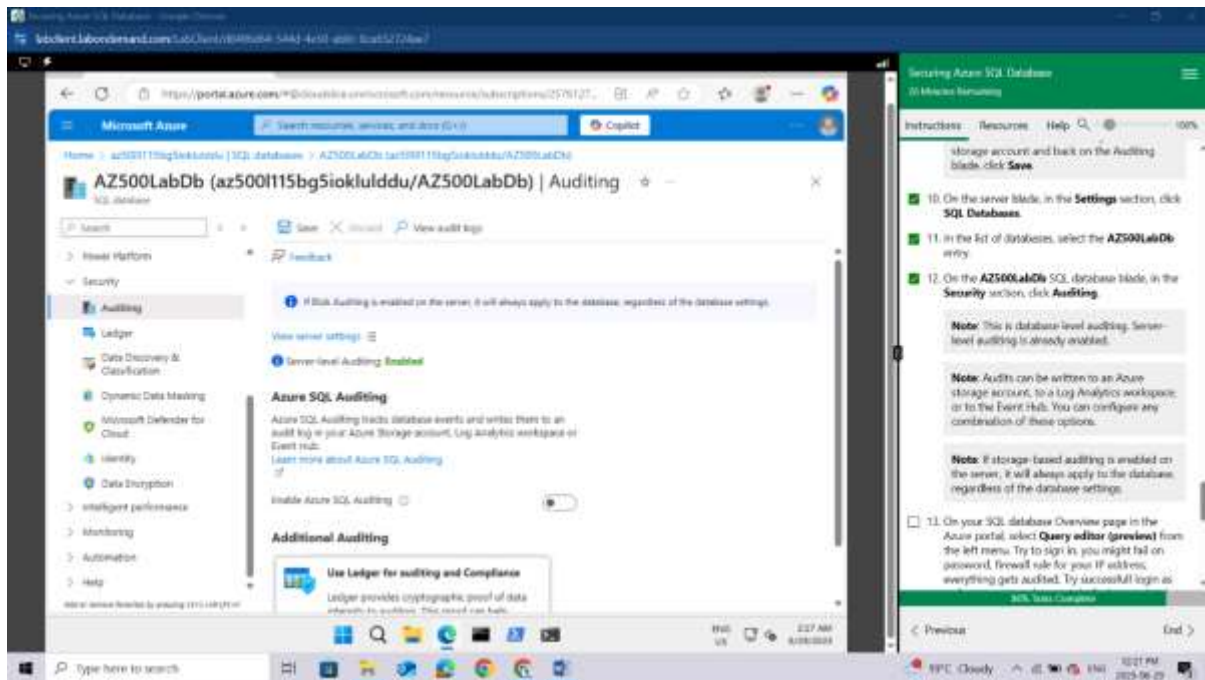


On the **AZ500LabDb** SQL database blade, in the **Security** section, click **Auditing**.

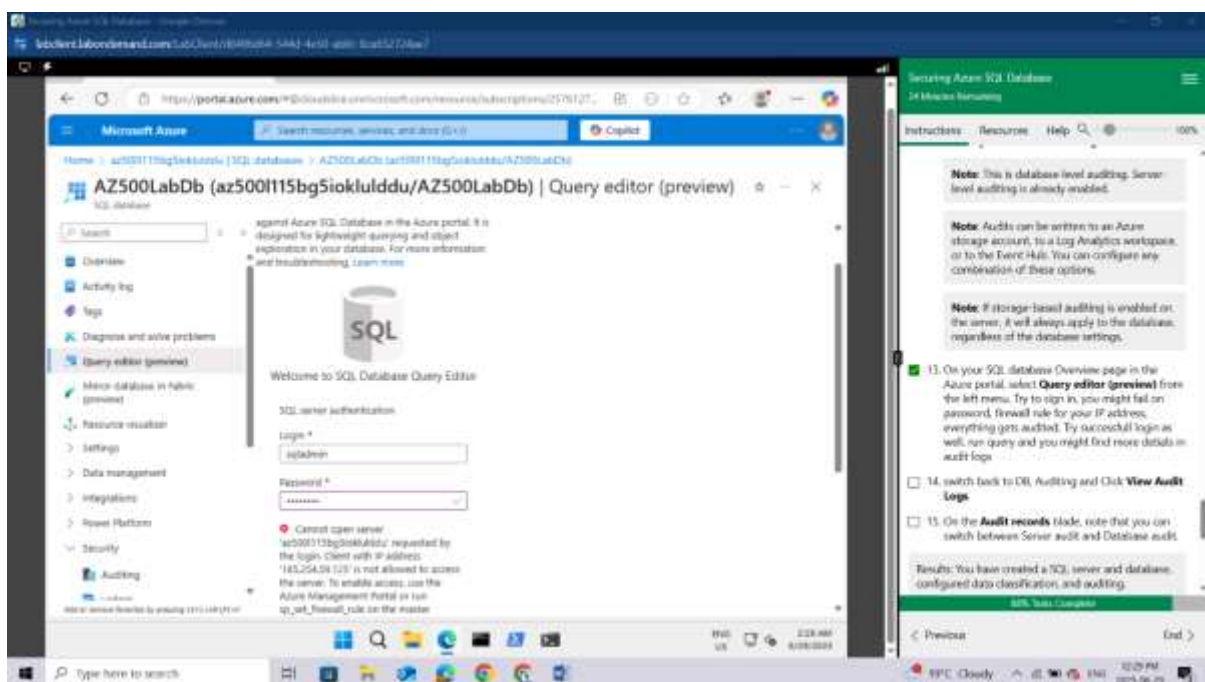
Note: This is database level auditing. Server-level auditing is already enabled.

Note: Audits can be written to an Azure storage account, to a Log Analytics workspace, or to the Event Hub. You can configure any combination of these options.

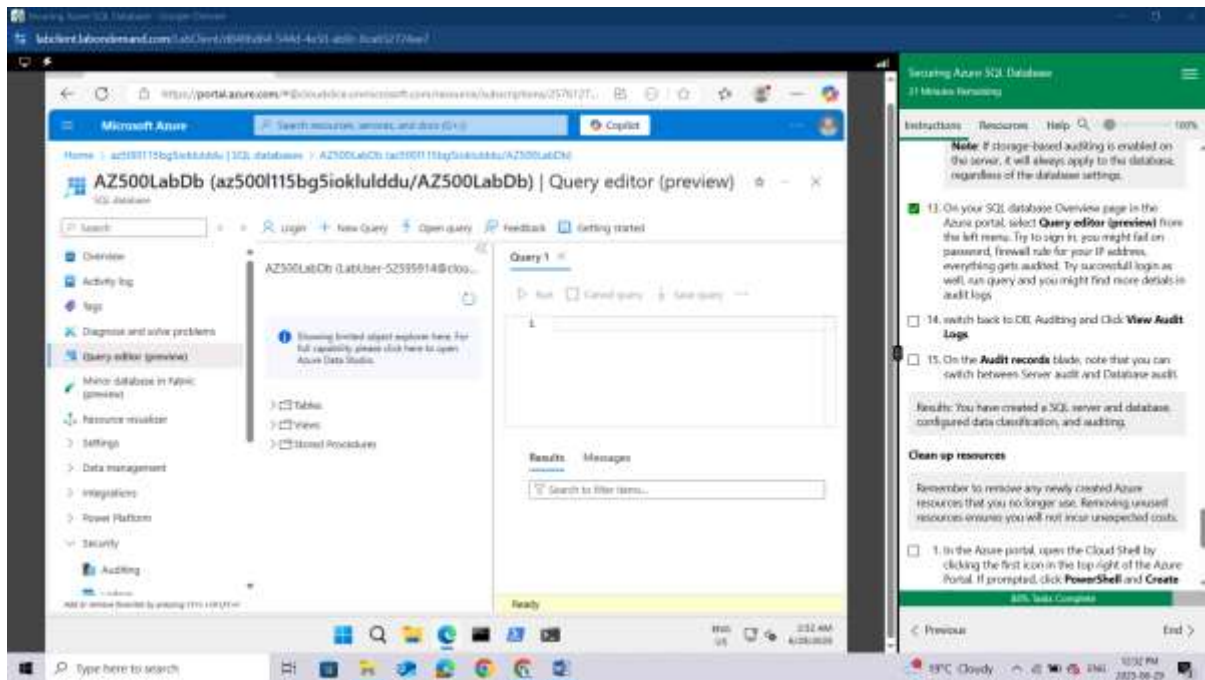
Note: If storage-based auditing is enabled on the server, it will always apply to the database, regardless of the database settings.



On your SQL database Overview page in the Azure portal, select **Query editor (preview)** from the left menu. Try to sign in, you might fail on password, firewall rule for your IP address, everything gets audited.

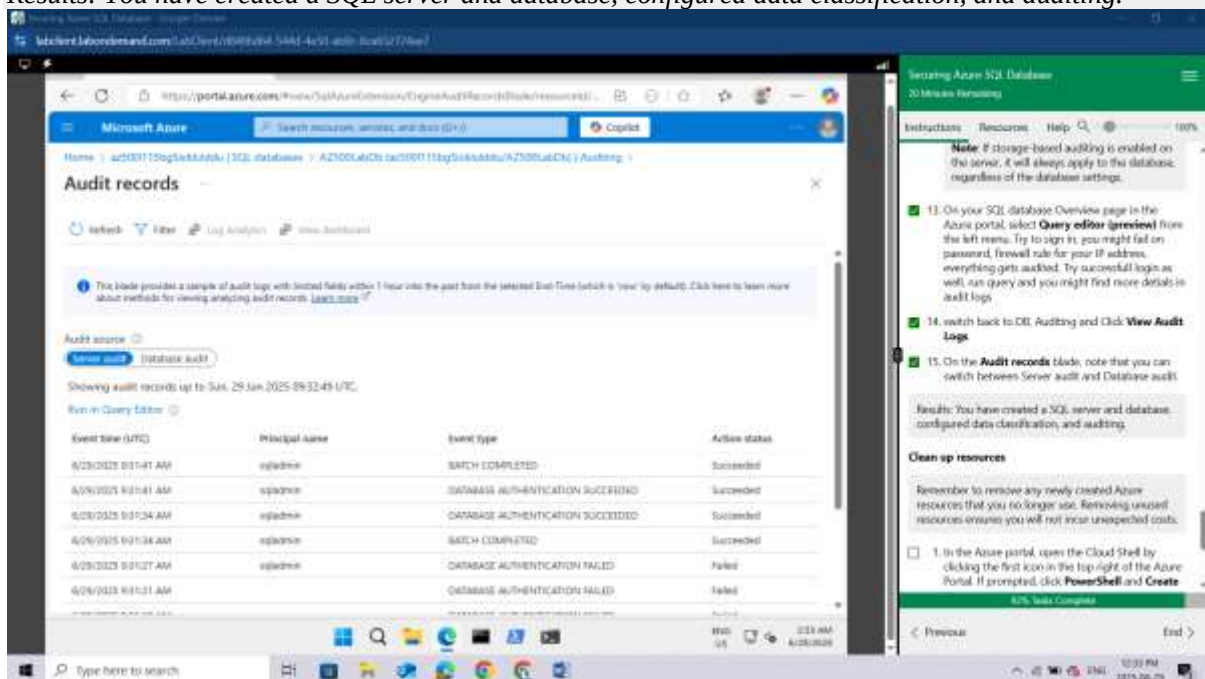


Try successful login as well, run query and you might find more details in audit logs.



Switch back to DB, Auditing and Click **View Audit Logs**. On the **Audit records** blade, note that you can switch between Server audit and Database audit.

Results: You have created a SQL server and database, configured data classification, and auditing.

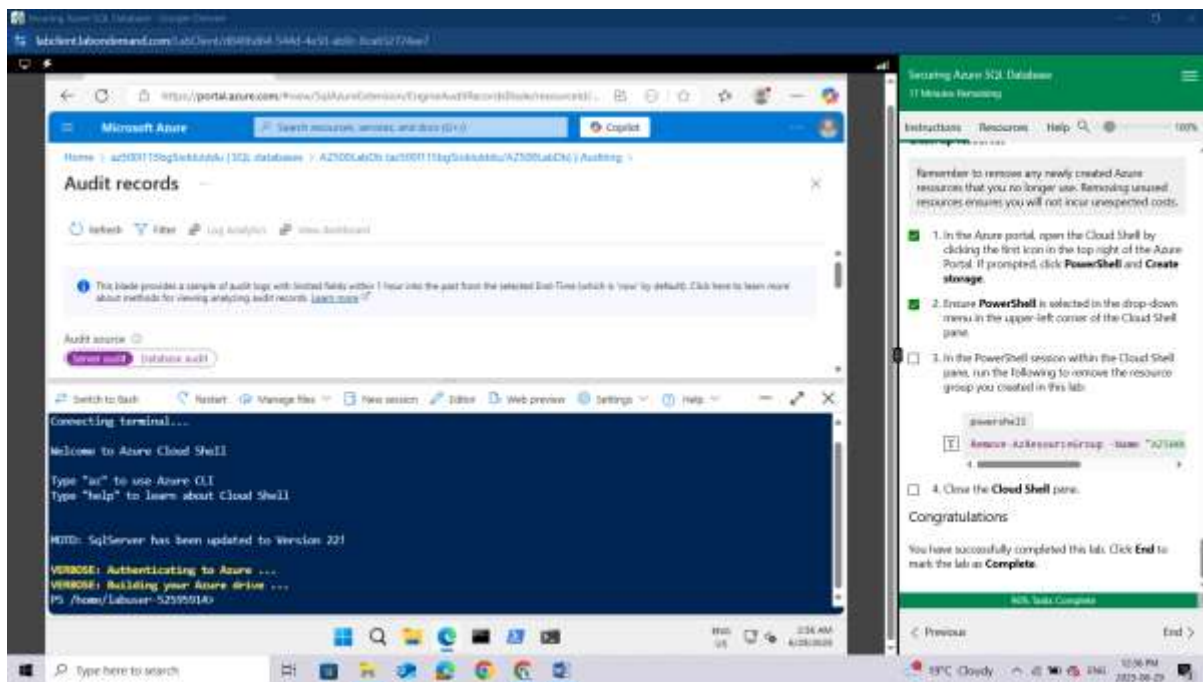


Clean up resources

Remember to remove any newly created Azure resources that you no longer use. Removing unused resources ensures you will not incur unexpected costs.

In the Azure portal, open the Cloud Shell by clicking the first icon in the top right of the Azure Portal. If prompted, click **PowerShell** and **Create storage**.

Ensure **PowerShell** is selected in the drop-down menu in the upper-left corner of the Cloud Shell pane.

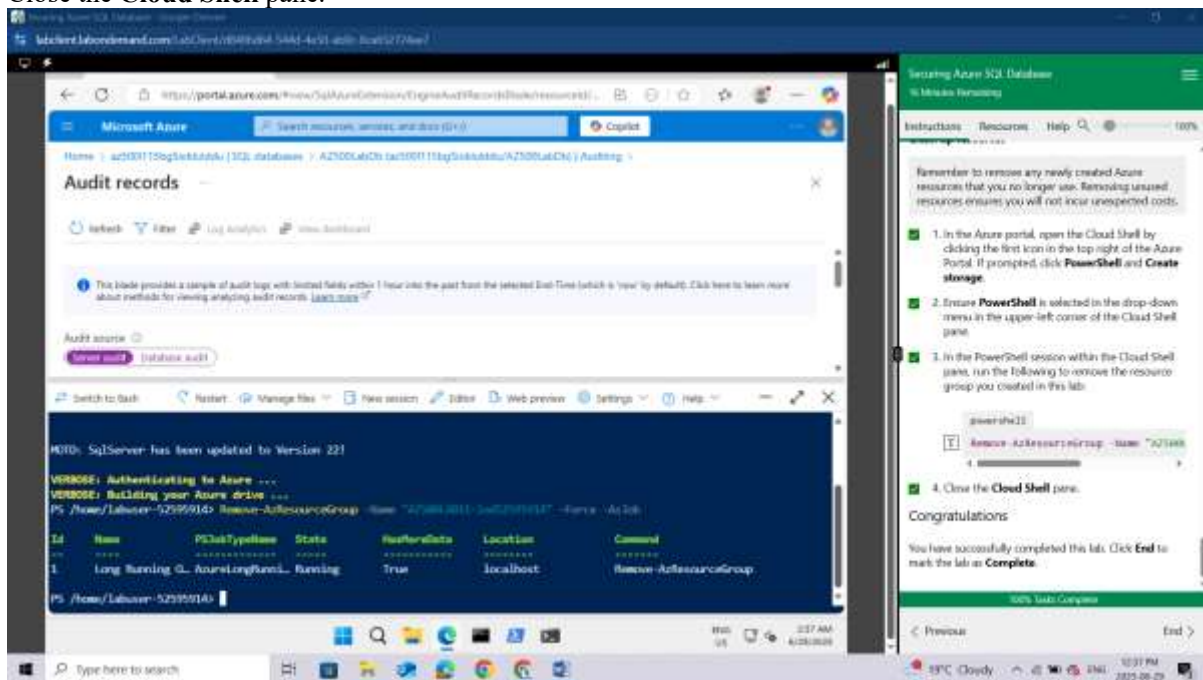


In the PowerShell session within the Cloud Shell pane, run the following to remove the resource group you created in this lab:

powershell

`Remove-AzResourceGroup -Name "AZ500LAB11-lod52595914" -Force -AsJob`

Close the **Cloud Shell** pane.



Congratulations! You have successfully completed this lab. Click **End** to mark the lab as **Complete**.

Conclusion

In this lab, I completed a series of tasks focused on securing an Azure SQL Database as part of the AZ500 Lab 05. First, I **deployed an Azure SQL Database** and ensured it was properly configured. Next, I enabled **Advanced Data Protection**, which provides security against potential threats like **SQL injection** and **data exfiltration**. I then moved on to **Data Classification**, where I discovered and labeled sensitive data within the database, categorizing it as **Confidential** to enhance visibility and compliance. Lastly, I configured **Auditing** to track and log database-level and server-level activities,

allowing for effective monitoring and incident investigation. Through these tasks, I gained practical experience in applying critical security measures to protect data hosted in Azure SQL Databases.